

XCPC Templates

Skyclad Observer

Continue

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1 代数结构

1.1 么半群

```

1 // Template
2 // struct Monoid {
3 //     struct Type {
4 //         i64 v;
5 //     };
6 //     static Type e() {
7 //         return {};
8 //     }
9 //     static Type op(const Type& a, const Type& b) {
10 //         return {};
11 //     }
12 //     static Type from(const i64& x) {
13 //         return {x};
14 //     }
15 // };
16
17 template <typename T>
18 struct AddMono {
19     using Type = T;
20     static Type e() {
21         return {0};
22     }
23     static Type op(const Type& a, const Type& b) {
24         return {a + b};
25     }
26     static Type from(const i64& x) {
27         return {x};
28     }
29     static Type inv(const Type& a) {
30         return {-a};
31     }
32     static Type pow(const Type& a, i64 k) {
33         return a * k;
34     }
35 };
36
37 template <typename T>
38 struct MulMono {
39     using Type = T;
40     static Type e() {

```

```

41         return {1};
42     }
43     static Type op(const Type& a, const Type& b) {
44         return {a * b};
45     }
46     static Type from(const i64& x) {
47         return {x};
48     }
49     static Type inv(const Type& a) {
50         return T{1} / a;
51     }
52 };
53
54 template <typename T>
55 struct MinMono {
56     using Type = T;
57     static Type e() {
58         return {inf<T>};
59     }
60     static Type op(const Type& a, const Type& b) {
61         return {min(a, b)};
62     }
63     static Type from(const i64& x) {
64         return {x};
65     }
66     static Type pow(const Type& a, i64 k) {
67         return (k == 0 ? e() : a);
68     }
69 };
70
71 template <typename T>
72 struct MaxMono {
73     using Type = T;
74     static Type e() {
75         return {-inf<T>};
76     }
77     static Type op(const Type& a, const Type& b) {
78         return {max(a, b)};
79     }
80     static Type from(const i64& x) {
81         return {x};
82     }
83     static Type pow(const Type& a, i64 k) {

```

```

84     return (k == 0 ? e() : a);
85 }
86 };
87
88 template <typename T>
89 struct GcdMono {
90     using Type = T;
91     static Type e() {
92         return {0};
93     }
94     static Type op(const Type& a, const Type& b) {
95         return {gcd(a, b)};
96     }
97     static Type from(const i64& x) {
98         return {x};
99     }
100    static Type pow(const Type& a, i64 k) {
101        return (k == 0 ? e() : a);
102    }
103 };
104
105 template <typename T>
106 struct LcmMono {
107     using Type = T;
108     static Type e() {
109         return {1};
110     }
111     static Type op(const Type& a, const Type& b) {
112         return {lcm(a, b)};
113     }
114     static Type from(const i64& x) {
115         return {x};
116     }
117     static Type pow(const Type& a, i64 k) {
118         return (k == 0 ? e() : a);
119     }
120 };
121
122 template <typename T>
123 struct XorMono {
124     using Type = T;
125     static Type e() {
126         return {0};

```

```

127     }
128     static Type op(const Type& a, const Type& b) {
129         return {a ^ b};
130     }
131     static Type inv(const Type& a) {
132         return {a};
133     }
134     static Type power(const Type& a, i64 k) {
135         return (k & 1) ? a : e();
136     }
137 };

```

1.2 幺半群作用

```

1 // Template
2 // struct ActMono {
3 //     using Mono = Mono;
4 //     using Tag = Tag;
5 //     using X = Mono::Type;
6 //     using Y = Tag::Type;
7 //     static X act(const X& x, const Y& y, int len) {
8 //         return {};
9 //     }
10 // };
11
12 template <typename T>
13 struct SumAdd {
14     using Mono = AddMono<T>;
15     using Tag = AddMono<T>;
16     using X = Mono::Type;
17     using Y = Tag::Type;
18     static X act(const X& x, const Y& y, int len) {
19         return {x + y * len};
20     }
21 };
22
23 template <typename T>
24 struct SumMul {
25     using Mono = AddMono<T>;
26     using Tag = MulMono<T>;
27     using X = Mono::Type;
28     using Y = Tag::Type;
29     static X act(const X& x, const Y& y, int len) {
30         return {x * y};

```

```

31     }
32 };
33
34 template <typename T>
35 struct MinAdd {
36     using Mono = MinMono<T>;
37     using Tag = AddMono<T>;
38     using X = Mono::Type;
39     using Y = Tag::Type;
40     static X act(const X& x, const Y& y, int len) {
41         return {x + y};
42     }
43 };
44
45 template <typename T>
46 struct MaxAdd {
47     using Mono = MaxMono<T>;
48     using Tag = AddMono<T>;
49     using X = Mono::Type;
50     using Y = Tag::Type;
51     static X act(const X& x, const Y& y, int len) {
52         return {x + y};
53     }
54 };

```

1.3 半群快速幂

```

1 template <typename Mono>
2 typename Mono::Type power(typename Mono::Type a, i64 b) {
3     if constexpr (requires { Mono::pow(a, b); }) {
4         return Mono::pow(a, b);
5     } else {
6         auto res = Mono::e();
7         while (b > 0) {
8             if (b & 1) {
9                 res = Mono::op(res, a);
10            }
11            a = Mono::op(a, a);
12            b >>= 1;
13        }
14        return res;
15    }
16 }

```

2 模运算

2.1 Barrett 模乘

```

1 class Barrett {
2 public:
3     constexpr Barrett(uint32_t Mod) : mod_(Mod), mu_(static_cast<uint64_t>(-1) /
Mod + 1) {}
4
5     constexpr uint32_t mod() const {
6         return mod_;
7     }
8
9     constexpr uint32_t multiply(uint32_t a, uint32_t b) const {
10         uint64_t z = static_cast<uint64_t>(a) * b;
11         uint64_t q = static_cast<uint64_t>((static_cast<__uint128_t>(z) * mu_) >>
64);
12         uint64_t y = q * mod_;
13         return static_cast<uint32_t>(z - y + (z < y ? mod_ : 0));
14     }
15 private:
16     uint32_t mod_;
17     uint64_t mu_;
18 };
19

```

2.2 运算

```

1 i64 mul(i64 a, i64 b, i64 m) {
2     return static_cast<i128>(a) * b % m;
3 }
4
5 i64 power(i64 a, i64 b, i64 m) {
6     i64 res = 1 % m;
7     while (b > 0) {
8         if (b & 1) {
9             res = mul(res, a, m);
10        }
11        a = mul(a, a, m);
12        b >>= 1;
13    }
14    return res;
15 }
16
17 template <typename T, std::integral U>
18 constexpr T power(T a, U b) {

```

```

19  assert(b >= 0);
20  T res{1};
21  while (b) {
22      if (b & 1) {
23          res *= a;
24      }
25      b >>= 1;
26      a *= a;
27  }
28  return res;
29 }

```

2.3 模数类

```

1  template <u32 P>
2  struct Mint {
3      u32 v;
4      Mint(i64 v_ = 0) : v((v_ %= P) < 0 ? v_ + P : v_) {}
5      Mint inv() const {
6          i64 x, y;
7          exgcd(v, P, x, y);
8          return x;
9      }
10     Mint& operator+=(Mint rhs) & {
11         v += rhs.v;
12         if (v >= P) v -= P;
13         return *this;
14     }
15     Mint& operator-=(Mint rhs) & {
16         v -= rhs.v;
17         if (v >= P) v += P;
18         return *this;
19     }
20     Mint& operator*=(Mint rhs) & {
21         v = u64(v) * rhs.v % P;
22         return *this;
23     }
24     Mint& operator/=(Mint rhs) & {
25         return *this *= rhs.inv();
26     }
27     friend Mint operator-(Mint rhs) { return rhs.v ? Mint(P - rhs.v) : rhs; }
28     friend Mint operator+(Mint lhs, Mint rhs) { return lhs += rhs; }
29     friend Mint operator-(Mint lhs, Mint rhs) { return lhs -= rhs; }
30     friend Mint operator*(Mint lhs, Mint rhs) { return lhs *= rhs; }

```

```

31     friend Mint operator/(Mint lhs, Mint rhs) { return lhs /= rhs; }
32     friend bool operator==(Mint lhs, Mint rhs) { return lhs.v == rhs.v; }
33     friend bool operator!=(Mint lhs, Mint rhs) { return lhs.v != rhs.v; }
34     friend istream& operator>>(istream& is, Mint& v) { i64 x; is >> x; v = x;
return is; }
35     friend ostream& operator<<(ostream& os, const Mint& v) { return os << v.v; }
36 };
37
38 constexpr u32 P = 998244353;
39 using Fp = Mint<P>;

```

2.4 all inverse

```

1  template <typename Fp>
2  vector<Fp> allInverse(const vector<Fp>& a) {
3      for (auto& v : a) {
4          assert(v != 0);
5      }
6      int n = a.size();
7      vector<Fp> res(n + 1);
8      res[0] = Fp{1};
9      for (int i = 0; i < n; i++) {
10         res[i + 1] = res[i] * a[i];
11     }
12     Fp inv = res[n].inverse();
13     res.pop_back();
14     for (int i = n - 1; i >= 0; i--) {
15         res[i] *= inv;
16         inv *= a[i];
17     }
18     return res;
19 }

```

3 图论

3.1 Base

```

1  template <typename T>
2  struct Edge {
3      int u, v;
4      T w;
5      Edge() {}
6      Edge(int u_, int v_, T w_) : u(u_), v(v_), w(w_) {}
7  };
8

```

```

9 template <typename T>
10 struct Graph {
11     vector<Edge<T>> edges;
12     vector<vector<pair<int, T>>> adj;
13     int n;
14
15     Graph() : n(0) {}
16     Graph(int n_) : n(n_) {
17         adj.resize(n);
18     }
19
20     void addEdge(int u, int v, T w = {0}) {
21         edges.emplace_back(u, v, w);
22         adj[u].emplace_back(v, w);
23     }
24 };

```

3.2 最短路

3.2.1 Dijkstra

```

1 template <typename T>
2 pair<vector<T>, vector<int>> dijkstra(vector<vector<pair<int, T>>>& adj, int s) {
3     int n = adj.size();
4     vector<T> dist(n, inf<T>);
5     vector<int> pre(n, -1);
6     using P = pair<T, int>;
7     priority_queue<P, vector<P>, greater<P>> pq;
8
9     dist[s] = 0;
10    pq.emplace(0, s);
11
12    while (!pq.empty()) {
13        auto [d, u] = pq.top();
14        pq.pop();
15        if (d != dist[u]) {
16            continue;
17        }
18        for (auto& [v, w] : adj[u]) {
19            if (dist[v] > dist[u] + w) {
20                dist[v] = dist[u] + w;
21                pre[v] = u;
22                pq.emplace(dist[v], v);
23            }
24        }
25    }
26 }

```

```

25 }
26
27     return {dist, pre};
28 }
29
30 template <typename T>
31 pair<vector<T>, vector<int>> dijkstraDense(vector<vector<pair<int, T>>>& adj, int s) {
32     int n = adj.size();
33     vector<T> dist(n, inf<T>);
34     vector<int> pre(n, -1);
35     vector<int> done(n, false);
36
37     dist[s] = 0;
38
39     for (int _ = 0; _ < n; _++) {
40         int u = -1;
41         for (int v = 0; v < n; v++) {
42             if (!done[v] && (u == -1 || dist[v] < dist[u])) {
43                 u = v;
44             }
45         }
46         if (u == -1 || dist[u] == inf<T>) {
47             break;
48         }
49         done[u] = true;
50         for (auto [v, w] : adj[u]) {
51             if (dist[v] > dist[u] + w) {
52                 dist[v] = dist[u] + w;
53                 pre[v] = u;
54             }
55         }
56     }
57
58     return {dist, pre};
59 }
60
61 template <typename T>
62 tuple<vector<T>, vector<int>, vector<int>> dijkstra(vector<vector<pair<int, T>>>& adj, vector<int>& s) {
63     int n = adj.size();
64     vector<T> dist(n, inf<T>);
65     vector<int> pre(n, -1), root(n, -1);
66     using P = pair<T, int>;

```

```

67 priority_queue<P, vector<P>, greater<P>> pq;
68
69 for (auto& u : s) {
70     dist[u] = 0;
71     root[u] = u;
72     pq.emplace(0, u);
73 }
74
75 while (!pq.empty()) {
76     auto [d, u] = pq.top();
77     pq.pop();
78     if (d != dist[u]) {
79         continue;
80     }
81     for (auto& [v, w] : adj[u]) {
82         if (dist[v] > dist[u] + w) {
83             dist[v] = dist[u] + w;
84             pre[v] = u;
85             root[v] = root[u];
86             pq.emplace(dist[v], v);
87         }
88     }
89 }
90
91 return {dist, pre, root};
92 }

```

3.2.2 01-BFS

```

1 template <typename T>
2 pair<vector<T>, vector<int>> bfs_01(vector<vector<pair<int, T>>& adj, int s) {
3     int n = adj.size();
4     vector<T> dist(n, inf<T>);
5     vector<int> pre(n, -1);
6     deque<int> que;
7
8     dist[s] = 0;
9     que.push_front(s);
10
11     while (!que.empty()) {
12         auto u = que.front();
13         que.pop_front();
14         for (auto& [v, w] : adj[u]) {
15             if (dist[v] > dist[u] + w) {
16                 dist[v] = dist[u] + w;

```

```

17         pre[v] = u;
18         if (w == 0) {
19             que.push_front(v);
20         } else {
21             que.push_back(v);
22         }
23     }
24 }
25
26
27 return {dist, pre};
28 }
29
30 template <typename T>
31 tuple<vector<T>, vector<int>, vector<int>> bfs_01(vector<vector<pair<int, T>>& adj, vector<int>& s) {
32     int n = adj.size();
33     vector<T> dist(n, inf<T>);
34     vector<int> pre(n, -1), root(n, -1);
35     deque<int> que;
36
37     for (auto& u : s) {
38         dist[u] = 0;
39         root[u] = u;
40         que.push_front(u);
41     }
42
43     while (!que.empty()) {
44         auto u = que.front();
45         que.pop_front();
46         for (auto& [v, w] : adj[u]) {
47             if (dist[v] > dist[u] + w) {
48                 dist[v] = dist[u] + w;
49                 pre[v] = u;
50                 root[v] = root[u];
51                 if (w == 0) {
52                     que.push_front(v);
53                 } else {
54                     que.push_back(v);
55                 }
56             }
57         }
58     }

```

```

59     return {dist, pre, root};
60 }
61

```

3.2.3 Bellman-Ford

```

1  template <typename T>
2  pair<vector<T>, vector<int>> bellmanFord(vector<vector<pair<int, T>>& adj, int s)
3  {
4      int n = adj.size();
5      vector<T> dist(n, inf<T>);
6      vector<int> pre(n, -1);
7
8      dist[s] = 0;
9
10     for (int round = 1;; round++) {
11         bool upd = false;
12         for (int u = 0; u < n; u++) {
13             if (dist[u] == inf<T>) {
14                 continue;
15             }
16             for (auto& [v, w] : adj[u]) {
17                 if (dist[v] > dist[u] + w) {
18                     dist[v] = dist[u] + w;
19                     pre[v] = u;
20                     upd = true;
21                     if (round >= n) {
22                         return {{}, {}};
23                     }
24                 }
25             }
26         }
27         if (!upd) {
28             break;
29         }
30     }
31     return {dist, pre};
32 }
33

```

3.2.4 SPFA

```

1  template <typename T>
2  pair<vector<T>, vector<int>> spfa(vector<vector<pair<int, T>>& adj, int s) {
3      int n = adj.size();

```

```

4      vector<T> dist(n, inf<T>);
5      vector<int> pre(n, -1);
6
7      dist[s] = 0;
8
9      queue<int> q;
10     vector<char> inq(n, false);
11     vector<int> cnt(n, 0);
12     q.push(s);
13     inq[s] = true;
14
15     while (!q.empty()) {
16         int u = q.front();
17         q.pop();
18         inq[u] = false;
19         for (auto& [v, w] : adj[u]) {
20             if (dist[v] > dist[u] + w) {
21                 dist[v] = dist[u] + w;
22                 pre[v] = u;
23                 if (!inq[v]) {
24                     q.push(v);
25                     inq[v] = true;
26                     cnt[v]++;
27                     if (cnt[v] >= n) {
28                         return {{}, {}};
29                     }
30                 }
31             }
32         }
33     }
34
35     return {dist, pre};
36 }

```

3.2.5 Floyd

```

1  template <typename T>
2  void floyd(vector<vector<T>>& dist) {
3      int n = dist.size();
4      for (int k = 0; k < n; k++) {
5          for (int i = 0; i < n; i++) {
6              for (int j = 0; j < n; j++) {
7                  if (dist[i][k] < inf<T> && dist[k][j] < inf<T>) {
8                      dist[i][j] = min(dist[i][j], dist[i][k] + dist[k][j]);

```

```

9         }
10    }
11 }
12 }
13 }
14
15 template <typename T>
16 vector<vector<T>> floyd(vector<vector<pair<int, T>>& adj) {
17     int n = adj.size();
18     vector dist(n, vector<T>(n, inf<T>));
19     for (int u = 0; u < n; u++) {
20         dist[u][u] = 0;
21         for (auto& [v, w] : adj[u]) {
22             dist[u][v] = min(dist[u][v], w);
23         }
24     }
25     floyd(dist);
26     return dist;
27 }
28

```

3.2.6 路径重建

```

1 vector<int> restorePath(vector<int>& pre, int t) {
2     vector<int> path = {t};
3     while (pre[path.back()] != -1) {
4         path.push_back(pre[path.back()]);
5     }
6     reverse(path.begin(), path.end());
7     return path;
8 }

```

3.3 拓扑排序

```

1 template <typename T>
2 vector<int> toposort(vector<vector<pair<int, T>>& adj) {
3     int n = adj.size();
4     vector<int> topo, deg(n, 0);
5     for (int u = 0; u < n; u++) {
6         for (auto& [v, w] : adj[u]) {
7             deg[v]++;
8         }
9     }
10    for (int u = 0; u < n; u++) {
11        if (deg[u] == 0) {

```

```

12            topo.push_back(u);
13        }
14    }
15    for (int i = 0; i < int(topo.size()); i++) {
16        int u = topo[i];
17        for (auto& [v, w] : adj[u]) {
18            if (--deg[v] == 0) {
19                topo.push_back(v);
20            }
21        }
22    }
23    return (int(topo.size()) != n ? vector<int>() : topo);
24 }
25
26 template <typename T>
27 vector<int> lexMinToposort(vector<vector<pair<int, T>>& adj) {
28     int n = adj.size();
29     vector<int> topo, deg(n, 0);
30     for (int u = 0; u < n; u++) {
31         for (auto& [v, w] : adj[u]) {
32             deg[v]++;
33         }
34     }
35     priority_queue<int, vector<int>, greater<int>> pq;
36     for (int u = 0; u < n; u++) {
37         if (deg[u] == 0) {
38             pq.push(u);
39         }
40     }
41     while (!pq.empty()) {
42         int u = pq.top();
43         pq.pop();
44         topo.push_back(u);
45         for (auto& [v, w] : adj[u]) {
46             if (--deg[v] == 0) {
47                 pq.push(v);
48             }
49         }
50     }
51     return (int(topo.size()) != n ? vector<int>() : topo);
52 }

```

3.4 欧拉通路/回路

给定起点 s ，还需要满足所有有边的顶点属于同一个连通分量，且 s 不能是孤立点。

无向图：

- 奇度点的数量一定是偶数，因此不会出现只有 1 个奇度点的情况。
- 有 0 个奇度点时，存在从任意有边顶点 s 出发的欧拉回路。
- 有 2 个奇度点时， s 必须是其中一个，存在从 s 出发的欧拉通路。
- 奇度点超过 2 个时不存在欧拉通路。

有向图：

- 若所有顶点满足 $\text{in} = \text{out}$ ，存在从任意有边顶点 s 出发的欧拉回路。
- 若恰有一个顶点满足 $\text{out} = \text{in} + 1$ ，它必须是 s ；同时恰有一个顶点满足 $\text{in} = \text{out} + 1$ ，存在从 s 出发的欧拉通路。
- 其他情况不存在欧拉通路。

```

1 // return:
2 // 0 - isn't Eulerian path
3 // 1 - is Eulerian path
4 // 2 - is Eulerian circuit
5 template <typename T>
6 int isEulerian(vector<vector<pair<int, T>>>& adj, bool directed = true, int s = -1)
7 {
8     int n = adj.size();
9     // in and out are only for directed graph
10    vector<int> in(n, 0), out(n, 0), deg(n, 0);
11    DSU dsu(n);
12    for (int u = 0; u < n; u++) {
13        for (auto& [v, w] : adj[u]) {
14            dsu.merge(u, v);
15            if (directed) {
16                out[u]++;
17                in[v]++;
18                deg[v]++;
19            }
20            deg[u]++;
21        }
22    }
23    int root = -1;
24    for (int u = 0; u < n; u++) {
25        if (deg[u] == 0) {
26            continue;
27        }
28        if (root == -1) {
29            root = u;

```

```

30    } else if (!dsu.same(root, u)) {
31        return 0;
32    }
33 }
34
35 if (s != -1 && root != -1 && !dsu.same(s, root)) {
36     return 0;
37 }
38
39 if (directed) {
40     int diff = 0;
41     for (int u = 0; u < n; u++) {
42         diff += abs(out[u] - in[u]);
43     }
44     if (diff == 0) {
45         return 2;
46     } else if (diff == 2) {
47         if (s == -1) {
48             return 1;
49         } else {
50             return (out[s] == in[s] + 1);
51         }
52     } else {
53         return 0;
54     }
55 } else {
56     int odd = 0;
57     for (int u = 0; u < n; u++) {
58         odd += (deg[u] & 1);
59     }
60     if (odd == 0) {
61         return 2;
62     } else if (odd == 2) {
63         if (s == -1) {
64             return 1;
65         } else {
66             return (deg[s] & 1);
67         }
68     } else {
69         return 0;
70     }
71 }
72 }

```

```

73
74 template <typename T>
75 bool isEulerianPath(vector<vector<pair<int, T>>& adj, bool directed = true, int s
76 = -1) {
77     return isEulerian(adj, directed, s) >= 1;
78 }
79 template <typename T>
80 bool isEulerianCircuit(vector<vector<pair<int, T>>& adj, bool directed = true, int
81 s = -1) {
82     return isEulerian(adj, directed, s) == 2;
83 }
84 template <typename T>
85 pair<vector<int>, vector<Edge<T>>> findEulerian(vector<vector<pair<int, T>>& adj,
86 int s = -1) {
87     // TODO
88 }

```

3.5 最小生成树

3.5.1 Kruskal

```

1 template <typename T>
2 pair<vector<Edge<T>>, T> kruskal(vector<Edge<T>>& edges, int n) {
3     T cost = 0;
4     vector<int> ord(edges.size());
5     iota(ord.begin(), ord.end(), 0);
6     sort(ord.begin(), ord.end(), [&](int i, int j) {
7         return edges[i].w < edges[j].w;
8     });
9
10    DSU dsu(n);
11    vector<Edge<T>> res;
12
13    for (auto& id : ord) {
14        auto& e = edges[id];
15        if (dsu.merge(e.u, e.v)) {
16            res.push_back(e);
17            cost += e.w;
18        }
19    }
20
21    return {res, cost};
22 }
23

```

3.5.2 Prim

```

1 template <typename T, typename F>
2 vector<Edge<T>> prim(int n, F cost) {
3     vector<Edge<T>> res;
4     vector<T> weight(n, inf<T>);
5     vector<int> pre(n);
6
7     auto add = [&](int u) -> void {
8         for (int v = 0; v < n; v++) {
9             if (pre[v] == -1) {
10                 continue;
11             }
12             T w = cost(u, v);
13             if (weight[v] > w) {
14                 weight[v] = w;
15                 pre[v] = u;
16             }
17         }
18         weight[u] = inf<T>;
19         pre[u] = -1;
20     };
21
22    add(0);
23    for (int i = 0; i < n - 1; i++) {
24        int u = min_element(weight.begin(), weight.end()) - weight.begin();
25        res.emplace_back(pre[u], u, weight[u]);
26        add(u);
27    }
28
29    return res;
30 }
31
32 template <typename T>
33 vector<Edge<T>> prim(vector<vector<pair<int, T>>& adj) {
34     int n = adj.size();
35     vector<Edge<T>> res;
36     vector<T> weight(n, inf<T>);
37     vector<int> pre(n);
38
39    auto add = [&](int u) -> void {
40        for (auto& [v, w] : adj[u]) {
41            if (pre[v] == -1) {
42                continue;
43            }

```

```

44     if (weight[v] > w) {
45         weight[v] = w;
46         pre[v] = u;
47     }
48 }
49 weight[u] = inf<T>;
50 pre[u] = -1;
51 };
52
53 add(0);
54 for (int i = 0; i < n - 1; i++) {
55     int u = min_element(weight.begin(), weight.end()) - weight.begin();
56     res.emplace_back(pre[u], u, weight[u]);
57     add(u);
58 }
59
60 return res;
61 }

```

3.5.3 Borůvka

```

1 template <typename T>
2 pair<vector<Edge<T>>, T> boruvka(vector<Edge<T>>& edges, int n) {
3     T cost = 0;
4     vector<Edge<T>> res;
5     DSU dsu(n);
6     int cnt = n;
7
8     while (cnt > 1) {
9         vector<int> best(n, -1);
10        int m = edges.size();
11
12        for (int i = 0; i < m; i++) {
13            auto& e = edges[i];
14            int u = dsu.find(e.u);
15            int v = dsu.find(e.v);
16            if (u == v) {
17                continue;
18            }
19            if (best[u] == -1 || e.w < edges[best[u]].w) {
20                best[u] = i;
21            }
22            if (best[v] == -1 || e.w < edges[best[v]].w) {
23                best[v] = i;

```

```

24        }
25    }
26
27    bool upd = false;
28    for (int u = 0; u < n; u++) {
29        if (best[u] == -1) {
30            continue;
31        }
32        auto& e = edges[best[u]];
33        if (dsu.merge(e.u, e.v)) {
34            res.push_back(e);
35            cost += e.w;
36            cnt--;
37            upd = true;
38        }
39    }
40    if (!upd) {
41        break;
42    }
43 }
44
45 return {res, cost};
46 }

```

3.5.4 Kruskal 重构树

```

1 template <typename T>
2 struct KruskalTree {
3     int n, root;
4     vector<i64> val;
5     vector<Edge<T>> edges;
6     vector<vector<int>> adj;
7
8     KruskalTree(int n_) {
9         init(n_);
10    }
11
12    void init(int n_) {
13        n = n_;
14        root = n - 1;
15        val.assign(2 * n - 1, 0);
16        adj.assign(2 * n - 1, vector<int>{});
17        edges.clear();
18    }

```

```

19 void addEdge(int u, int v, i64 w) {
20     edges.emplace_back(u, v, w);
21 }
22
23 void build() {
24     DSU dsu(2 * n - 1);
25     vector<int> ord(2 * n - 1);
26     iota(ord.begin(), ord.end(), 0);
27     sort(ord.begin(), ord.end(), [&](int x, int y) {
28         return edges[x].w < edges[y].w;
29     });
30
31     for (auto& id : ord) {
32         auto& e = edges[id];
33         int u = e.u, v = e.v;
34         i64 w = e.w;
35
36         if (!dsu.same(u, v)) {
37             u = dsu.find(u);
38             v = dsu.find(v);
39             val[++root] = w;
40
41             adj[root].push_back(u);
42             adj[root].push_back(v);
43             adj[u].push_back(root);
44             adj[v].push_back(root);
45
46             dsu.f[u] = root;
47             dsu.f[v] = root;
48             dsu.siz[root] = dsu.siz[u] + dsu.siz[v];
49
50             if (root == 2 * n - 2) {
51                 return;
52             }
53         }
54     }
55 }
56
57 };

```

3.6 连通性

3.6.1 强连通分量

```

1 struct SCC {
2     int n;

```

```

3     vector<vector<int>>> adj;
4     vector<int> stk;
5     vector<int> dfn, low, bel;
6     int cur, cnt;
7
8     SCC() {}
9     SCC(int n_) {
10         init(n_);
11     }
12
13     void init(int n_) {
14         n = n_;
15         adj.assign(n, {});
16         dfn.assign(n, -1);
17         low.resize(n);
18         bel.assign(n, -1);
19         stk.clear();
20         cur = cnt = 0;
21     }
22
23     void addEdge(int u, int v) {
24         adj[u].push_back(v);
25     }
26
27     void dfs(int u) {
28         dfn[u] = low[u] = cur++;
29         stk.push_back(u);
30
31         for (auto& v : adj[u]) {
32             if (dfn[v] == -1) {
33                 dfs(v);
34                 low[u] = min(low[u], low[v]);
35             } else if (bel[v] == -1) {
36                 low[u] = min(low[u], dfn[v]);
37             }
38         }
39
40         if (dfn[u] == low[u]) {
41             int y;
42             do {
43                 y = stk.back();
44                 bel[y] = cnt;
45                 stk.pop_back();

```

```

46         } while (y != u);
47         cnt++;
48     }
49 }
50
51 vector<int> work() {
52     for (int i = 0; i < n; i++) {
53         if (dfn[i] == -1) {
54             dfs(i);
55         }
56     }
57     for (auto& x : bel) {    // 把逆拓扑序转成拓扑序
58         x = cnt - 1 - x;
59     }
60     return bel;
61 }
62 };

```

3.6.2 边双连通分量

```

1 struct EBCC {
2     int n;
3     vector<vector<pair<int, int>>> adj;
4     vector<int> stk;
5     vector<int> dfn, low, bel;
6     int cur, cnt, ecnt;
7
8     EBCC() {}
9     EBCC(int n_) {
10         init(n_);
11     }
12
13     void init(int n_) {
14         n = n_;
15         adj.assign(n, {});
16         dfn.assign(n, -1);
17         low.resize(n);
18         bel.assign(n, -1);
19         stk.clear();
20         cur = cnt = ecnt = 0;
21     }
22
23     void addEdge(int u, int v) {
24         adj[u].emplace_back(v, ecnt);

```

```

25         adj[v].emplace_back(u, ecnt);
26         ecnt++;
27     }
28
29     void dfs(int u, int p) {
30         dfn[u] = low[u] = cur++;
31         stk.push_back(u);
32
33         for (auto& [v, eid] : adj[u]) {
34             if (eid == p) {
35                 continue;
36             }
37             if (dfn[v] == -1) {
38                 dfs(v, eid);
39                 low[u] = min(low[u], low[v]);
40             } else {
41                 low[u] = min(low[u], dfn[v]);
42             }
43         }
44
45         if (dfn[u] == low[u]) {
46             int y;
47             do {
48                 y = stk.back();
49                 bel[y] = cnt;
50                 stk.pop_back();
51             } while (y != u);
52             cnt++;
53         }
54     }
55
56     vector<int> work() {
57         for (int i = 0; i < n; i++) {
58             if (dfn[i] == -1) {
59                 dfs(i, -1);
60             }
61         }
62         return bel;
63     }
64 };

```

3.6.3 点双连通分量

```

1 struct VBCC {

```

```

2  int n;
3  vector<pair<int, int>> edges;
4  vector<vector<pair<int, int>>> adj;
5  vector<int> stk;
6  vector<int> dfn, low, cut;
7  vector<vector<int>> comp;
8  int cur;
9
10 VBCC() {}
11 VBCC(int n_) {
12     init(n_);
13 }
14
15 void init(int n_) {
16     n = n_;
17     adj.assign(n, {});
18     edges.clear();
19     dfn.assign(n, -1);
20     low.resize(n);
21     stk.clear();
22     comp.clear();
23     cut.assign(n, false);
24     cur = 0;
25 }
26
27 void addEdge(int u, int v) {
28     adj[u].emplace_back(v, edges.size());
29     adj[v].emplace_back(u, edges.size());
30     edges.emplace_back(u, v);
31 }
32
33 void dfs(int u, int p) {
34     dfn[u] = low[u] = cur++;
35     int child = 0;
36     for (auto& [v, eid] : adj[u]) {
37         if (eid == p) {
38             continue;
39         }
40         if (dfn[v] == -1) {
41             child++;
42             stk.push_back(eid);
43             dfs(v, eid);
44             low[u] = min(low[u], low[v]);

```

```

45         if (low[v] >= dfn[u]) {
46             if (p != -1) {
47                 cut[u] = true;
48             }
49             vector<int> b;
50             while (true) {
51                 int e = stk.back();
52                 stk.pop_back();
53                 b.push_back(e);
54                 if (e == eid) {
55                     break;
56                 }
57             }
58             comp.push_back(b);
59         }
60     } else if (dfn[v] < dfn[u]) {
61         stk.push_back(eid);
62         low[u] = min(low[u], dfn[v]);
63     }
64 }
65 if (p == -1 && child >= 2) {
66     cut[u] = true;
67 }
68 }
69
70 void work() {
71     for (int i = 0; i < n; i++) {
72         if (dfn[i] == -1) {
73             dfs(i, -1);
74         }
75     }
76 }
77 };

```

3.6.4 圆方树

```

1 struct BlockCutTree {
2     int n;
3     vector<pair<int, int>> edges;
4     vector<vector<pair<int, int>>> adj;
5     vector<int> stk;
6     vector<int> dfn, low, cut;
7     vector<vector<int>> comp;
8     vector<vector<int>> tree;

```

```

9   int cur;
10  int tot;

11
12  BlockCutTree() {}
13  BlockCutTree(int n_) {
14      init(n_);
15  }
16
17  void init(int n_) {
18      n = n_;
19      adj.assign(n, {});
20      edges.clear();
21      dfn.assign(n, -1);
22      low.resize(n);
23      stk.clear();
24      comp.clear();
25      cut.assign(n, false);
26      tree.assign(n, {});
27      tot = n;
28      cur = 0;
29  }
30
31  void addEdge(int u, int v) {
32      adj[u].emplace_back(v, edges.size());
33      adj[v].emplace_back(u, edges.size());
34      edges.emplace_back(u, v);
35  }
36
37  void addBlock(vector<int>& b) {
38      int x = tot++;
39      tree.emplace_back();
40      vector<int> pts;
41      for (auto& e : b) {
42          auto [u, v] = edges[e];
43          pts.push_back(u);
44          pts.push_back(v);
45      }
46      sort(pts.begin(), pts.end());
47      pts.erase(unique(pts.begin(), pts.end(), pts.end()), pts.end());
48      for (int u : pts) {
49          tree[u].push_back(x);
50          tree[x].push_back(u);
51      }

```

```

52  }
53
54  void dfs(int u, int p) {
55      dfn[u] = low[u] = cur++;
56      int child = 0;
57      for (auto& [v, eid] : adj[u]) {
58          if (eid == p) {
59              continue;
60          }
61          if (dfn[v] == -1) {
62              child++;
63              stk.push_back(eid);
64              dfs(v, eid);
65              low[u] = min(low[u], low[v]);
66              if (low[v] >= dfn[u]) {
67                  if (p != -1) {
68                      cut[u] = true;
69                  }
70                  vector<int> b;
71                  while (true) {
72                      int e = stk.back();
73                      stk.pop_back();
74                      b.push_back(e);
75                      if (e == eid) {
76                          break;
77                      }
78                  }
79                  comp.push_back(b);
80                  addBlock(b);
81              }
82              } else if (dfn[v] < dfn[u]) {
83                  stk.push_back(eid);
84                  low[u] = min(low[u], dfn[v]);
85              }
86          }
87          if (p == -1 && child >= 2) {
88              cut[u] = true;
89          }
90      }
91
92  void build() {
93      for (int i = 0; i < n; i++) {
94          if (dfn[i] == -1) {

```

```

95     dfs(i, -1);
96 }
97 }
98 }
99 };

```

3.7 树链剖分

```

1 struct HLD {
2     int n;
3     vector<int> siz, top, dep, fa, in, out, seq;
4     vector<vector<int>> adj;
5     int cur;
6
7     HLD() {}
8     HLD(int n_) {
9         n = n_;
10        siz.resize(n);
11        top.resize(n);
12        dep.resize(n);
13        fa.resize(n);
14        in.resize(n);
15        out.resize(n);
16        seq.resize(n);
17        cur = 0;
18        adj.assign(n, {});
19    }
20
21    void addEdge(int u, int v) {
22        adj[u].push_back(v);
23        adj[v].push_back(u);
24    }
25
26    void work(int root = 0) {
27        top[root] = root;
28        dep[root] = 0;
29        fa[root] = -1;
30        dfs1(root);
31        dfs2(root);
32    }
33
34    void dfs1(int u) {
35        if (fa[u] != -1) {
36            adj[u].erase(find(adj[u].begin(), adj[u].end(), fa[u]));

```

```

37        }
38
39        siz[u] = 1;
40        for (auto& v : adj[u]) {
41            fa[v] = u;
42            dep[v] = dep[u] + 1;
43            dfs1(v);
44            siz[u] += siz[v];
45            if (siz[v] > siz[adj[u][0]]) {
46                swap(v, adj[u][0]);
47            }
48        }
49    }
50
51    void dfs2(int u) {
52        in[u] = cur++;
53        seq[in[u]] = u;
54        for (auto& v : adj[u]) {
55            top[v] = (v == adj[u][0] ? top[u] : v);
56            dfs2(v);
57        }
58        out[u] = cur;
59    }
60
61    int lca(int u, int v) {
62        while (top[u] != top[v]) {
63            if (dep[top[u]] > dep[top[v]]) {
64                u = fa[top[u]];
65            } else {
66                v = fa[top[v]];
67            }
68        }
69        return dep[u] < dep[v] ? u : v;
70    }
71
72    int dist(int u, int v) {
73        return dep[u] + dep[v] - 2 * dep[lca(u, v)];
74    }
75
76    int jump(int u, int k) {
77        if (dep[u] < k) {
78            return -1;
79        }

```

```

80     int d = dep[u] - k;
81     while (dep[top[u]] > d) {
82         u = fa[top[u]];
83     }
84     return seq[in[u] - dep[u] + d];
85 }
86
87 bool isAncestor(int u, int v) {
88     return in[u] <= in[v] && in[v] < out[u];
89 }
90
91 int rootedParent(int u, int v) {
92     swap(u, v);
93     if (u == v) {
94         return u;
95     }
96     if (!isAncestor(u, v)) {
97         return fa[u];
98     }
99     auto it = upper_bound(adj[u].begin(), adj[u].end(), v, [&](int x, int y)
100 { return in[x] < in[y]; }) - 1;
101     return *it;
102 }
103
104 int rootedSize(int u, int v) {
105     if (u == v) {
106         return n;
107     }
108     if (!isAncestor(v, u)) {
109         return siz[v];
110     }
111     return n - siz[rootedParent(u, v)];
112 }
113
114 int rootedLca(int a, int b, int c) {
115     return lca(a, b) ^ lca(b, c) ^ lca(c, a);
116 }
117
118 };

```

3.8 LCA

```

1 struct LCA {
2     int logn, cur;
3     vector<int> dep, dfn;

```

```

4     vector<vector<int>>> fa;
5
6     LCA(vector<vector<int>>& adj, int root = 0) {
7         init(adj, root);
8     }
9
10    void init(vector<vector<int>>& adj, int root) {
11        int n = adj.size();
12        cur = 0;
13        logn = __lg(n);
14        fa.assign(n, vector<int>(logn + 1, -1));
15        dep.assign(n, 0);
16        dfn.assign(n, -1);
17
18        auto dfs = [&](auto&& self, int u, int p) -> void {
19            fa[u][0] = p;
20            dfn[u] = cur++;
21            for (int i = 1; i <= logn; i++) {
22                fa[u][i] = fa[fa[u][i - 1]][i - 1];
23            }
24            for (auto& v : adj[u]) {
25                if (v == p) {
26                    continue;
27                }
28                dep[v] = dep[u] + 1;
29                self(self, v, u);
30            }
31        };
32
33        dfs(dfs, root, root);
34
35        for (int i = n - 1; i >= 0; i--) {
36            if (fa[i][0] == -1) {
37                dfs(dfs, i, i);
38            }
39        }
40    }
41
42    int get(int u, int v) {
43        if (dep[u] < dep[v]) {
44            swap(u, v);
45        }
46        for (int i = logn; i >= 0; i--) {

```

```

47     if (dep[fa[u][i]] >= dep[v]) {
48         u = fa[u][i];
49     }
50 }
51 if (u == v) {
52     return u;
53 }
54 for (int i = logn; i >= 0; i--) {
55     if (fa[u][i] != fa[v][i]) {
56         u = fa[u][i];
57         v = fa[v][i];
58     }
59 }
60 return fa[u][0];
61 }
62
63 int dist(int u, int v) {
64     int res = dep[u] + dep[v] - 2 * dep[get(u, v)];
65     return res;
66 }
67 };
68

```

3.9 树上倍增

```

1  template <typename Mono>
2  struct TreeLifting {
3      using X = typename Mono::Type;
4
5      vector<int> dep;
6      BinaryLifting<Mono> lift;
7
8      TreeLifting() {}
9
10     template<typename T>
11     TreeLifting(vector<vector<pair<int, T>>& adj, int root = 0) {
12         int n = adj.size();
13         dep.assign(n, 0);
14         lift.init(n);
15         auto dfs = [&](auto&& self, int u, int p) -> void {
16             for (auto& [v, w] : adj[u]) {
17                 if (v == p) {
18                     continue;
19                 }

```

```

20         dep[v] = dep[u] + 1;
21         lift.add(v, u, Mono::from(w));
22         self(self, v, u);
23     }
24 };
25 lift.add(root, -1, Mono::e());
26 dfs(dfs, root, -1);
27 lift.build();
28 }
29
30 int lca(int u, int v) {
31     if (dep[u] < dep[v]) {
32         swap(u, v);
33     }
34     u = lift.jump(u, dep[u] - dep[v]);
35     if (u == v) {
36         return u;
37     }
38     for (int k = lift.logn; k >= 0; k--) {
39         if (lift.nxt[k][u] != lift.nxt[k][v]) {
40             u = lift.nxt[k][u];
41             v = lift.nxt[k][v];
42         }
43     }
44     return lift.nxt[0][u];
45 }
46
47 pair<int, X> query(int u, int v) {
48     int l = lca(u, v);
49     X res = Mono::op(lift.query(u, dep[u] - dep[l]).second, lift.query(v,
50 dep[v] - dep[l]).second);
51     return {l, res};
52 }
53 };

```

3.10 虚树

```

1  template <typename T>
2  struct VirtualTree {
3      LCA& lca;
4      vector<int> nodes;
5      vector<vector<pair<int, T>>& adj;
6
7      VirtualTree(LCA& lca) : lca(lca) {}

```

```

8
9  pair<vector<int>&, vector<vector<pair<int, T>>>&> build(const vector<int>& h) {
10     nodes = h;
11     auto cmp = [&](int u, int v) { return lca.dfn[u] < lca.dfn[v]; };
12     sort(nodes.begin(), nodes.end(), cmp);
13     int m = nodes.size();
14     for (int i = 1; i < m; i++) {
15         nodes.push_back(lca.get(nodes[i - 1], nodes[i]));
16     }
17     sort(nodes.begin(), nodes.end(), cmp);
18     nodes.erase(unique(nodes.begin(), nodes.end()), nodes.end());
19     int n = nodes.size();
20     adj.resize(n);
21     for (auto& e : adj) {
22         e.clear();
23     }
24     for (int i = 0; i < n - 1; i++) {
25         int p = lca.get(nodes[i], nodes[i + 1]);
26         int u = lower_bound(nodes.begin(), nodes.end(), p, cmp) -
nodes.begin();
27         int v = i + 1;
28         T w = lca.dep[nodes[v]] - lca.dep[p];
29         adj[u].emplace_back(v, w);
30         adj[v].emplace_back(u, w);
31     }
32
33     return {nodes, adj};
34 }
35 };

```

3.11 2-SAT

addClause(u, f, v, g) 加入一个析取子句:

$$(x_u = f) \vee (x_v = g).$$

这里 x_i 是第 i 个布尔变量, 变量编号从 0 开始。f 和 g 决定文字的正负: true 表示正文字 x_i , false 表示负文字 $\neg x_i$ 。

例如: addClause(3, true, 5, false) 表示

$$x_3 \vee \neg x_5.$$

一个子句 $(A \vee B)$ 等价于两条蕴含边:

$$(\neg A) \rightarrow B, (\neg B) \rightarrow A.$$

所以 addClause(u, f, v, g) 实际加入:

$$(x_u = \neg f) \rightarrow (x_v = g), (x_v = \neg g) \rightarrow (x_u = f).$$

图中的编号约定是: $2i$ 表示变量 x_i 取假值, $2i+1$ 表示变量 x_i 取真值。因此第一条边的起点是第一个文字的否定, 终点是第二个文字本身; 另一条边同理。

```

1 struct TwoSat {
2     int n;
3     SCC scc;
4     vector<int> ans;
5
6     TwoSat(int n_) : n(n_), scc(2 * n), ans(n) {}
7
8     // add clause: (x_u = f) ∨ (x_v = g)
9     void addClause(int u, bool f, int v, bool g) {
10         scc.addEdge(2 * u + !f, 2 * v + g);
11         scc.addEdge(2 * v + !g, 2 * u + f);
12     }
13
14     // (x_u = f)
15     void set(int u, bool f) {
16         addClause(u, f, u, f);
17     }
18
19     // (x_u = f) -> (x_v = g)
20     void implies(int u, int f, int v, int g) {
21         addClause(u, !f, v, g);
22     }
23
24     pair<bool, vector<int>> satisfiable() {
25         auto id = scc.work();
26         for (int i = 0; i < n; i++) {
27             if (id[2 * i] == id[2 * i + 1]) {
28                 return {false, {}};
29             }
30             ans[i] = id[2 * i] < id[2 * i + 1];
31         }
32         return {true, ans};
33     }
34 };

```

4 网络流

4.1 最大流最小割

```

1 template <typename T>
2 struct MaxFlow {
3     struct Edge_ {
4         int to;
5         T cap;
6         Edge_(int to_, T cap_) : to(to_), cap(cap_) {}

```

```

7   };
8
9   int n;
10  vector<Edge_> e;
11  vector<vector<int>> adj;
12  vector<int> cur, level;
13
14  MaxFlow() {}
15  MaxFlow(int n_) {
16      init(n_);
17  }
18
19  void init(int n_) {
20      n = n_;
21      e.clear();
22      adj.assign(n_, {});
23      cur.resize(n_);
24      level.resize(n_);
25  }
26
27  bool bfs(int s, int t) {
28      level.assign(n, -1);
29      queue<int> q;
30      level[s] = 0;
31      q.push(s);
32      while (!q.empty()) {
33          const int u = q.front();
34          q.pop();
35          for (auto& i : adj[u]) {
36              auto [v, c] = e[i];
37              if (c > 0 && level[v] == -1) {
38                  level[v] = level[u] + 1;
39                  if (v == t) {
40                      return true;
41                  }
42                  q.push(v);
43              }
44          }
45      }
46      return false;
47  }
48
49  T dfs(int u, int t, T f) {

```

```

50      if (u == t) {
51          return f;
52      }
53      auto r = f;
54      for (int& i = cur[u]; i < int(adj[u].size()); i++) {
55          const int& j = adj[u][i];
56          auto [v, c] = e[j];
57          if (c > 0 && level[v] == level[u] + 1) {
58              auto a = dfs(v, t, min(r, c));
59              e[j].cap -= a;
60              e[j ^ 1].cap += a;
61              r -= a;
62              if (r == 0) {
63                  return f;
64              }
65          }
66      }
67      return f - r;
68  }
69
70  void addEdge(int u, int v, T c) {
71      adj[u].push_back(e.size());
72      e.emplace_back(v, c);
73      adj[v].push_back(e.size());
74      e.emplace_back(u, 0);
75  }
76
77  T flow(int s, int t) {
78      T ans = 0;
79      while (bfs(s, t)) {
80          cur.assign(n, 0);
81          ans += dfs(s, t, numeric_limits<T>::max());
82      }
83      return ans;
84  }
85
86  vector<bool> minCut() {
87      vector<bool> c(n);
88      for (int i = 0; i < n; i++) {
89          c[i] = (level[i] != -1);
90      }
91      return c;
92  }

```

```

93 struct Edge {
94     int from;
95     int to;
96     T cap;
97     T flow;
98 };
99
100 vector<Edge> edges() {
101     vector<Edge> a;
102     for (int i = 0; i < e.size(); i += 2) {
103         Edge x;
104         x.from = e[i + 1].to;
105         x.to = e[i].to;
106         x.cap = e[i].cap + e[i + 1].cap;
107         x.flow = e[i + 1].cap;
108         a.push_back(x);
109     }
110     return a;
111 }
112 }
113 };

```

4.2 最小费用最大流

最小费用流的每次增广都要在残量网络上找最短路，而残量网络里有反向边、费用是负的，Dijkstra 直接用不了。势能数组 h 就是用来把边权「平移」成非负的。

做法是把每条边的费用换成约化费用

$$c'(u, v) = c(u, v) + h(u) - h(v),$$

并且每轮求完最短路后令 $h(v) \leftarrow h(v) + \text{dist}'(v)$ ，其中 dist' 是用约化费用算出的最短路。这样之后的约化费用依然非负——由三角不等式 $\text{dist}'(v) \leq \text{dist}'(u) + c'(u, v)$ 整理一下，正好就是新的 $c'(u, v) \geq 0$ 。

代价是路径长度被平移了常量 $h(s) - h(t)$ ，但它不影响谁是最短路。代码里每轮累加的是 $h(t)$ ：因为 $h(s)$ 恒为 0，更新后的 $h(t)$ 恰好等于原网络中 $s \rightarrow t$ 的真实最短费用。

初始势能默认全 0，这就要求**初始边权全部非负**；如果原图本来就有负费用边，得根据图的特性自行构建 h 或者直接先跑一遍 SPFA 得到初始 h ，再从参数传进来。

```

1 template<typename T>
2 struct MinCostFlow {
3     struct Edge_ {
4         int to;
5         T cap;
6         T cost;
7         Edge_(int to_, T cap_, T cost_) : to(to_), cap(cap_), cost(cost_) {}
8     };
9 };

```

```

10 int n;
11 vector<Edge_> e;
12 vector<vector<int>> adj;
13 vector<T> h, dist;
14 vector<int> pre;
15
16 bool dijkstra(int s, int t) {
17     dist.assign(n, numeric_limits<T>::max());
18     pre.assign(n, -1);
19     priority_queue<pair<T, int>, vector<pair<T, int>>, greater<pair<T, int>>>
20     que;
21     dist[s] = 0;
22     que.emplace(0, s);
23     while (!que.empty()) {
24         auto [d, u] = que.top();
25         que.pop();
26         if (dist[u] != d) {
27             continue;
28         }
29         for (auto& i : adj[u]) {
30             int v = e[i].to;
31             if (e[i].cap > 0 && dist[v] > d + h[u] - h[v] + e[i].cost) {
32                 dist[v] = d + h[u] - h[v] + e[i].cost;
33                 pre[v] = i;
34                 que.emplace(dist[v], v);
35             }
36         }
37     }
38     return dist[t] != numeric_limits<T>::max();
39 }
40
41 MinCostFlow() {}
42 MinCostFlow(int n_) {
43     init(n_);
44 }
45
46 void init(int n_) {
47     n = n_;
48     e.clear();
49     adj.assign(n, {});
50 }
51
52 void addEdge(int u, int v, T cap, T cost) {
53     adj[u].push_back(e.size());

```

```

53     e.emplace_back(v, cap, cost);
54     adj[v].push_back(e.size());
55     e.emplace_back(u, 0, -cost);
56 }
57
58 pair<T, T> flow(int s, int t, T need = numeric_limits<T>::max(), vector<T>
potential = {}) {
59     T flow = 0;
60     T cost = 0;
61     if (potential.empty()) {
62         potential.assign(n, 0);
63     }
64     h = potential;
65     while (flow < need && dijkstra(s, t)) {
66         for (int i = 0; i < n; ++i) {
67             h[i] += dist[i];
68         }
69         T aug = numeric_limits<T>::max();
70         for (int i = t; i != s; i = e[pre[i] ^ 1].to) {
71             aug = min(aug, e[pre[i]].cap);
72         }
73         for (int i = t; i != s; i = e[pre[i] ^ 1].to) {
74             e[pre[i]].cap -= aug;
75             e[pre[i] ^ 1].cap += aug;
76         }
77         flow += aug;
78         cost += aug * h[t];
79     }
80     return {flow, cost};
81 }
82
83 struct Edge {
84     int from;
85     int to;
86     T cap;
87     T cost;
88     T flow;
89 };
90
91 vector<Edge> edges() {
92     vector<Edge> a;
93     for (int i = 0; i < e.size(); i += 2) {
94         Edge x;

```

```

95         x.from = e[i + 1].to;
96         x.to = e[i].to;
97         x.cap = e[i].cap + e[i + 1].cap;
98         x.cost = e[i].cost;
99         x.flow = e[i + 1].cap;
100        a.push_back(x);
101    }
102    return a;
103 }
104 };

```

5 数论

5.1 扩展欧几里得算法

```

1 i64 exgcd(i64 a, i64 b, i64& x, i64& y) {
2     if (b == 0) {
3         x = 1, y = 0;
4         return a;
5     }
6     i64 d = exgcd(b, a % b, y, x);
7     y -= a / b * x;
8     return d;
9 }
10
11 i64 inv(i64 a, i64 m) {
12     i64 x, y;
13     i64 d = exgcd(a, m, x, y);
14     if (d != 1) {
15         return -1;
16     }
17     return (x % m + m) % m;
18 }
19
20 bool solveLinear(i64 a, i64 b, i64 c, i64& x, i64& y) {
21     i64 d = exgcd(a, b, x, y);
22     if (c % d != 0) {
23         return false;
24     }
25     i64 k = c / d;
26     x *= k;
27     y *= k;
28     return true;
29 }

```

```

30
31 pair<i64, i64> solveMod(i64 a, i64 b, i64 m) {
32     assert(m > 0);
33     b *= -1;
34     i64 x, y;
35     i64 g = exgcd(a, m, x, y);
36     if (g < 0) {
37         g *= -1;
38         x *= -1;
39         y *= -1;
40     }
41     if (b % g != 0) {
42         return {-1, -1};
43     }
44     x = x * (b / g) % (m / g);
45     if (x < 0) {
46         x += m / g;
47     }
48     return {x, m / g};
49 }
50

```

5.2 线性筛

```

1 vector<int> minp, primes;
2
3 void sieve(int n) {
4     minp.assign(n + 1, 0);
5     primes.clear();
6
7     for (int i = 2; i <= n; i++) {
8         if (minp[i] == 0) {
9             minp[i] = i;
10            primes.push_back(i);
11        }
12
13        for (auto p : primes) {
14            if (i * p > n) {
15                break;
16            }
17            minp[i * p] = p;
18            if (p == minp[i]) {
19                break;
20            }
21        }
22    }
23

```

```

21     }
22 }
23 }

```

5.3 莫比乌斯反演

莫比乌斯反演用来在「按约数求和」和「按倍数求和」之间互转。

$\mu(n)$ 的定义: $n = 1$ 时为 1; n 是 k 个互异素数之积时为 $(-1)^k$; 含平方因子时为 0。一切从这条恒等式出发:

$$\sum_{d|n} \mu(d) = [n = 1].$$

于是有两种方向的反演:

$$f(n) = \sum_{d|n} g(d) \Rightarrow g(n) = \sum_{d|n} \mu(d) f\left(\frac{n}{d}\right),$$

$$f(n) = \sum_{n|d} g(d) \Rightarrow g(n) = \sum_{n|d} \mu\left(\frac{d}{n}\right) f(d).$$

实际做题更常用的是「换条件」的用法: 把 $[\gcd(i, j) = 1]$ 换成 $\sum_{d|\gcd(i, j)} \mu(d)$, 再交换求和顺序, 条件就从 $\gcd$ 挪到了 d :

$$\sum_{i=1}^n \sum_{j=1}^m [\gcd(i, j) = 1] = \sum_{d=1}^{\min(n, m)} \mu(d) \left\lfloor \frac{n}{d} \right\rfloor \left\lfloor \frac{m}{d} \right\rfloor.$$

右边只要 μ 的前缀和就能整除分块, μ 用下面的 `sieve()` 线性筛出来。

```

1 vector<int> minp, primes, mu;
2
3 void sieve(int n) {
4     minp.assign(n + 1, 0);
5     mu.assign(n + 1, 0);
6     primes.clear();
7     mu[1] = 1;
8
9     for (int i = 2; i <= n; i++) {
10        if (minp[i] == 0) {
11            minp[i] = i;
12            mu[i] = -1;
13            primes.push_back(i);
14        }
15
16        for (auto p : primes) {
17            if (i * p > n) {
18                break;
19            }
20            minp[i * p] = p;
21            if (p == minp[i]) {
22                mu[i * p] = 0;
23            }
24        }
25    }
26

```

```

23         break;
24     } else {
25         mu[i * p] = -mu[i];
26     }
27 }
28 }
29 }
30

```

5.4 欧拉函数

$$\varphi(n) = n \prod_{i=1}^k \left(1 - \frac{1}{p_i}\right).$$

两条常用性质：它是积性的，即 $\gcd(a, b) = 1$ 时 $\varphi(ab) = \varphi(a)\varphi(b)$ ；并且 $\sum_{d|n} \varphi(d) = n$ ，也即 $\varphi(n) = \sum_{d|n} \mu(d) \frac{n}{d}$ 。

欧拉定理： $\gcd(a, m) = 1$ 时 $a^{\varphi(m)} \equiv 1 \pmod{m}$ 。竞赛里更常用的是扩展欧拉定理，它给指数降幂—— $b \geq \varphi(m)$ 时对任意 a 都有

$$a^b \equiv a^{b \bmod \varphi(m) + \varphi(m)} \pmod{m}.$$

线性筛求 φ 的递推：素数的 $\varphi(p) = p - 1$ ；对 $i \cdot p$ ，若 $p \nmid i$ 则 $\varphi(ip) = p\varphi(i)$ ，否则 $\varphi(ip) = (p - 1)\varphi(i)$ 。见下面的 `sieve()`；只求单个数用 `Phi()`，复杂度 $O(\sqrt{n})$ 。

```

1 vector<int> minp, primes, phi;
2
3 // 线性筛（用于求解全部数的欧拉函数）
4 void sieve(int n) {
5     minp.assign(n + 1, 0);
6     phi.assign(n + 1, 0);
7     primes.clear();
8     phi[1] = 1;
9
10    for (int i = 2; i <= n; i++) {
11        if (minp[i] == 0) {
12            minp[i] = i;
13            primes.push_back(i);
14            phi[i] = i - 1;
15        }
16
17        for (auto p : primes) {
18            if (i * p > n) {
19                break;
20            }
21            minp[i * p] = p;
22            if (p == minp[i]) {
23                phi[i * p] = phi[i] * p;

```

```

24         break;
25     } else {
26         phi[i * p] = phi[i] * (p - 1);
27     }
28 }
29 }
30 }
31
32 // 求解单个数的欧拉函数
33 int Phi(int n) {
34     int res = n;
35     for (int i = 2; i * i <= n; i++) {
36         if (n % i == 0) {
37             while (n % i == 0) {
38                 n /= i;
39             }
40             res = res / i * (i - 1);
41         }
42     }
43     if (n > 1) {
44         res = res / n * (n - 1);
45     }
46     return res;
47 }
48

```

5.5 Miller-Rabin

```

1 bool isPrime(i64 n) {
2     if (n < 2) {
3         return false;
4     }
5     constexpr array<int, 10> smallPrimes = {2, 3, 5, 7, 11, 13, 17, 19, 23, 29};
6     for (auto& p : smallPrimes) {
7         if (n % p == 0) {
8             return n == p;
9         }
10    }
11    if (n < 31 * 31) {
12        return true;
13    }
14    constexpr array<int, 7> bases = {2, 325, 9375, 28178, 450775, 9780504,
15    1795265022};
16    int s = __builtin_ctzll(n - 1);

```

```

16 i64 d = (n - 1) >> s;
17 for (auto& a : bases) {
18     if (a % n == 0) {
19         continue;
20     }
21     i64 cur = power(a, d, n);
22     if (cur == 1) {
23         continue;
24     }
25     bool witness = true;
26     for (int r = 0; r < s; r++) {
27         if (cur == n - 1) {
28             witness = false;
29             break;
30         }
31         cur = mul(cur, cur, n);
32     }
33     if (witness) {
34         return false;
35     }
36 }
37 return true;
38 }

```

5.6 Pollard-Rho

```

1 static mt19937_64
2 rng(static_cast<u32>(chrono::steady_clock::now().time_since_epoch().count()));
3
4 i64 rho(i64 n, i64 c) {
5     auto f = [&](i64 x) {
6         return static_cast<i64>((static_cast<i128>(x) * x + c) % n);
7     };
8     i64 x = 1, y = 2, z = 1, q = 1;
9     i64 g = 1;
10    constexpr i64 m = 127;
11    for (i64 r = 1; g == 1; r <= 1) {
12        x = y;
13        for (i64 i = 0; i < r; i++) {
14            y = f(y);
15        }
16        for (i64 k = 0; k < r && g == 1; k += m) {
17            z = y;
18            for (i64 i = 0; i < min(m, r - k); i++) {

```

```

18                y = f(y);
19                q = mul(q, abs(x - y), n);
20            }
21            g = gcd(q, n);
22        }
23    }
24    if (g == n) {
25        do {
26            z = f(z);
27            g = gcd(abs(x - z), n);
28        } while (g == 1);
29    }
30    return g;
31 }
32
33 i64 primeFactor(i64 n) {
34     assert(n >= 2);
35     if (n % 2 == 0) {
36         return 2;
37     }
38     while (!isPrime(n)) {
39         uniform_int_distribution<i64> pick(1, n - 1);
40         n = rho(n, pick(rng));
41     }
42     return n;
43 }
44
45 vector<pair<i64, int>> factorize(i64 n) {
46     vector<pair<i64, int>> pf;
47     for (int p = 2; p < 100; p++) {
48         if (p * p > n) {
49             break;
50         }
51         if (n % p == 0) {
52             int e = 0;
53             do {
54                 n /= p;
55                 e++;
56             } while (n % p == 0);
57             pf.emplace_back(p, e);
58         }
59     }
60     while (n > 1) {

```

```

61     i64 p = primeFactor(n);
62     int e = 0;
63     do {
64         n /= p;
65         e++;
66     } while (n % p == 0);
67     pf.emplace_back(p, e);
68 }
69 sort(pf.begin(), pf.end());
70 return pf;
71 }

```

5.7 约数构造

```

1 vector<i64> buildDivisors(const vector<pair<i64, int>>& factors) {
2     vector<i64> divisors = {1};
3     for (auto& [p, e] : factors) {
4         int sz = divisors.size();
5         for (int i = 0; i < sz; i++) {
6             i64 x = divisors[i];
7             for (int j = 0; j < e; j++) {
8                 x *= p;
9                 divisors.push_back(x);
10            }
11        }
12    }
13    return divisors;
14 }

```

6 组合数学

6.1 组合数

```

1 template <typename T>
2 struct Combination {
3     int n;
4     vector<T> fact_, invfact_, inv_;
5
6     Combination() : n(0), fact_{1}, invfact_{1}, inv_{0} {}
7     Combination(int n_) : Combination() {
8         init(n_);
9     }
10
11     void init(int n_) {
12         if (n_ <= n) {

```

```

13         return;
14     }
15     fact_.resize(n_ + 1);
16     invfact_.resize(n_ + 1);
17     inv_.resize(n_ + 1);
18
19     for (int i = n + 1; i <= n_; i++) {
20         fact_[i] = fact_[i - 1] * i;
21     }
22     invfact_[n_] = fact_[n_].inv();
23     for (int i = n_; i > n; i--) {
24         invfact_[i - 1] = invfact_[i] * i;
25         inv_[i] = invfact_[i] * fact_[i - 1];
26     }
27     n = n_;
28 }
29
30 T fact(int k) {
31     if (k > n) {
32         init(2 * k);
33     }
34     return fact_[k];
35 }
36
37 T invfact(int k) {
38     if (k > n) {
39         init(2 * k);
40     }
41     return invfact_[k];
42 }
43
44 T inv(int k) {
45     if (k > n) {
46         init(2 * k);
47     }
48     return inv_[k];
49 }
50
51 T binom(int n, int m) {
52     if (n < m || m < 0) {
53         return 0;
54     }
55     return fact(n) * invfact(m) * invfact(n - m);

```

```

56 }
57
58 T perm(int n, int m) {
59     if (n < m || m < 0) {
60         return 0;
61     }
62     return fact(n) * invfact(n - m);
63 }
64
65 T multicombo(int n, int m) {
66     if (n < 0 || m < 0) {
67         return 0;
68     }
69     if (n == 0) {
70         return (m == 0);
71     }
72     return binom(n + m - 1, m);
73 }
74
75 T Catalan(int n) {
76     if (n < 0) {
77         return 0;
78     }
79     return binom(2 * n, n) * inv(n + 1);
80 }
81 };

```

6.2 康托展开

```

1 vector<int> toLehmer(vector<int>& p) {
2     int n = p.size();
3     Fenwick<AddMono<int>> fen(n);
4     vector<int> c(n);
5     for (int i = n - 1; i >= 0; i--) {
6         c[i] = fen.sum(p[i]);
7         fen.add(p[i], 1);
8     }
9     return c;
10 }
11
12 vector<int> fromLehmer(vector<int>& c) {
13     int n = c.size();
14     vector<int> p(n);
15     Fenwick<AddMono<int>> fen(n);

```

```

16     for (int i = 0; i < n; i++) {
17         fen.add(i, 1);
18     }
19     for (int i = 0; i < n; i++) {
20         p[i] = fen.kth(c[i]);
21         fen.add(p[i], -1);
22     }
23     return p;
24 }
25
26 i64 toRank(vector<int>& p) {
27     int n = p.size();
28     auto c = toLehmer(p);
29     i64 fac = 1, rk = 0;
30     for (int i = n - 1; i >= 0; i--) {
31         rk += c[i] * fac;
32         fac *= (n - i);
33     }
34     return rk;
35 }
36
37 vector<int> fromRank(int n, i64 rk) {
38     vector<int> c(n);
39     for (int i = 1; i <= n; i++) {
40         c[n - i] = rk % i;
41         rk /= i;
42     }
43     return fromLehmer(c);
44 }

```

6.3 逆序对

```

1 i64 invCount(vector<i64>& a, int l, int r, vector<i64>& tmp) {
2     if (r - l <= 1) return 0;
3
4     i64 res = 0;
5     int mid = (l + r) / 2;
6     res += invCount(a, l, mid, tmp);
7     res += invCount(a, mid, r, tmp);
8
9     int i = l, j = mid, k = l;
10    while (i < mid && j < r) {
11        if (a[i] <= a[j]) {
12            tmp[k++] = a[i++];

```

```

13     } else {
14         tmp[k++] = a[j++];
15         res += mid - i;
16     }
17 }
18 while (i < mid) {
19     tmp[k++] = a[i++];
20 }
21 while (j < r) {
22     tmp[k++] = a[j++];
23 }
24
25 for (i = l; i < r; i++) {
26     a[i] = tmp[i];
27 }
28
29 return res;
30 }
31

```

6.4 容斥原理

```

1 // cap(S) 表示 S 集合的交集大小
2
3 // 交集形式
4 template <typename T, typename F>
5 T ie(int n, F&& cap) {
6     T ans{};
7     for (int S = 1; S < (1 << n); S++) {
8         T cur = cap(S);
9         if (__builtin_popcount(S) & 1) {
10             ans += cur;
11         } else {
12             ans -= cur;
13         }
14     }
15     return ans;
16 }
17
18 // 补集形式
19 template <typename T, typename F>
20 T ieComp(int n, F&& cap) {
21     T ans{};
22     for (int S = 0; S < (1 << n); S++) {

```

```

23         T cur = cap(S);
24         if (__builtin_popcount(S) & 1) {
25             ans -= cur;
26         } else {
27             ans += cur;
28         }
29     }
30     return ans;
31 }
32
33 // 恰好满足 k 个性质的元素数量
34 template <typename T, typename F>
35 vector<T> ieExact(int n, F&& cap) {
36     vector<T> N(n + 1, T{});
37     for (int S = 0; S < (1 << n); S++) {
38         int k = __builtin_popcount(S);
39         N[k] += cap(S);
40     }
41
42     vector<T> binom(n + 2, T{}), E(n + 1, T{});
43     binom[0] = T{1};
44     for (int j = 0; j <= n; j++) {
45         for (int k = 0; k <= j; k++) {
46             T term = N[j] * binom[k];
47             if ((j - k) & 1) {
48                 E[k] -= term;
49             } else {
50                 E[k] += term;
51             }
52         }
53         for (int k = j + 1; k >= 1; k--) {
54             binom[k] += binom[k - 1];
55         }
56     }
57     return E;
58 }
59
60 // 至少满足 k 个
61 template <typename T, typename F>
62 vector<T> ieLeast(int n, F&& cap) {
63     auto E = ieExact<T>(n, cap);
64     for (int k = n - 1; k >= 0; k--) {
65         E[k] += E[k + 1];

```

```

66     }
67     return E;
68 }
69
70 // 至多满足 k 个
71 template <typename T, typename F>
72 vector<T> ieMost(int n, F&& cap) {
73     auto E = ieExact<T>(n, cap);
74     for (int k = 0; k < n; k++) {
75         E[k + 1] += E[k];
76     }
77     return E;
78 }

```

6.5 zeta 变换

```

1 // 子集和:  $f[\text{mask}] = \sum_{\text{sub} \subseteq \text{mask}} f[\text{sub}]$ 
2 // 方向: 小的 mask  $\rightarrow$  大的 mask ("吸收"不带位)
3 template <typename T>
4 void subsetZeta(vector<T>& f) {
5     int n = f.size();
6     for (int i = 1; i < n; i <= 1) {
7         for (int mask = 0; mask < n; mask++) {
8             if (mask & i) {
9                 f[mask] += f[mask ^ i];
10            }
11        }
12    }
13 }
14
15 // 子集和的逆
16 template <typename T>
17 void subsetMobius(vector<T>& f) {
18     int n = f.size();
19     for (int i = 1; i < n; i <= 1) {
20         for (int mask = 0; mask < n; mask++) {
21             if (mask & i) {
22                 f[mask] -= f[mask ^ i];
23            }
24        }
25    }
26 }
27

```

```

28 // 超集和:  $f[\text{mask}] = \sum_{\text{sup} \supseteq \text{mask}} f[\text{sup}]$ 
29 // 方向: 大的 mask  $\rightarrow$  小的 mask ("分发"给不带这一位的 mask)
30 template <typename T>
31 void supersetZeta(vector<T>& f) {
32     int n = f.size();
33     for (int i = 1; i < n; i <= 1) {
34         for (int mask = 0; mask < n; mask++) {
35             if (mask & i) {
36                 f[mask ^ i] += f[mask];
37            }
38        }
39    }
40 }
41
42 // 超集和的逆
43 template <typename T>
44 void supersetMobius(vector<T>& f) {
45     int n = f.size();
46     for (int i = 1; i < n; i <= 1) {
47         for (int mask = 0; mask < n; mask++) {
48             if (mask & i) {
49                 f[mask ^ i] -= f[mask];
50            }
51        }
52    }
53 }

```

7 数据结构

7.1 线段树

7.1.1 线段树

```

1 template <typename Mono>
2 struct SegmentTree {
3     using X = typename Mono::Type;
4
5     int n;
6     vector<X> info;
7
8     SegmentTree() : n(0) {}
9     SegmentTree(const vector<X>& init_) {
10         init(init_);
11     }
12

```

```

13 void init(const vector<X>& init_) {
14     n = init_.size();
15     int logn = __lg(n);
16     info.assign(4 << logn, Mono::e());
17
18     auto build = [&](auto&& self, int p, int l, int r) -> void {
19         if (r - l == 1) {
20             info[p] = init_[l];
21             return;
22         }
23         int mid = (l + r) / 2;
24         self(self, 2 * p, l, mid);
25         self(self, 2 * p + 1, mid, r);
26         pull(p);
27     };
28
29     build(build, 1, 0, n);
30 }
31
32 void pull(int p) {
33     info[p] = Mono::op(info[2 * p], info[2 * p + 1]);
34 }
35
36 void modify(int p, int l, int r, int x, const X& v) {
37     if (r - l == 1) {
38         info[p] = v;
39         return;
40     }
41     int mid = (l + r) / 2;
42     if (x < mid) {
43         modify(2 * p, l, mid, x, v);
44     } else {
45         modify(2 * p + 1, mid, r, x, v);
46     }
47     pull(p);
48 }
49
50 void modify(int p, const X& v) {
51     modify(1, 0, n, p, v);
52 }
53
54 X rangeQuery(int p, int l, int r, int x, int y) {
55     if (y <= l || r <= x) {

```

```

56     }
57     if (x <= l && r <= y) {
58         return info[p];
59     }
60     int mid = (l + r) / 2;
61     return Mono::op(rangeQuery(2 * p, l, mid, x, y), rangeQuery(2 * p + 1, mid,
r, x, y));
62 }
63 X rangeQuery(int l, int r) {
64     return rangeQuery(1, 0, n, l, r);
65 }
66
67 template <typename F>
68 int findFirst(int p, int l, int r, int x, int y, F&& pred) {
69     if (y <= l || r <= x) {
70         return -1;
71     }
72     if (x <= l && r <= y && !pred(info[p])) {
73         return -1;
74     }
75     if (r - l == 1) {
76         return l;
77     }
78     int mid = (l + r) / 2;
79     int res = findFirst(2 * p, l, mid, x, y, pred);
80     if (res == -1) {
81         res = findFirst(2 * p + 1, mid, r, x, y, pred);
82     }
83     return res;
84 }
85
86 template <typename F>
87 int findFirst(int l, int r, F&& pred) {
88     return findFirst(1, 0, n, l, r, pred);
89 }
90
91 template <typename F>
92 int findLast(int p, int l, int r, int x, int y, F&& pred) {
93     if (y <= l || r <= x) {
94         return -1;
95     }
96     if (x <= l && r <= y && !pred(info[p])) {
97         return -1;

```

```

98     if (r - l == 1) {
99         return l;
100     }
101     int mid = (l + r) / 2;
102     int res = findLast(2 * p + 1, mid, r, x, y, pred);
103     if (res == -1) {
104         res = findLast(2 * p, l, mid, x, y, pred);
105     }
106     return res;
107 }
108 template <typename F>
109 int findLast(int l, int r, F&& pred) {
110     return findLast(1, 0, n, l, r, pred);
111 }
112 };

```

7.1.2 懒标记线段树

```

1 template <typename ActMono>
2 struct LazySegmentTree {
3     using Mono = typename ActMono::Mono;
4     using Tag = typename ActMono::Tag;
5     using X = typename ActMono::X;
6     using Y = typename ActMono::Y;
7
8     int n;
9     vector<X> info;
10    vector<Y> tag;
11
12    LazySegmentTree() : n(0) {}
13    LazySegmentTree(const vector<X>& init_) {
14        init(init_);
15    }
16
17    void init(const vector<X>& init_) {
18        n = init_.size();
19        int logn = __lg(n);
20        info.assign(4 << logn, Mono::e());
21        tag.assign(4 << logn, Tag::e());
22
23        auto build = [&](auto&& self, int p, int l, int r) -> void {
24            if (r - l == 1) {
25                info[p] = init_[l];
26                return;

```

```

27        }
28        int mid = (l + r) / 2;
29        self(self, 2 * p, l, mid);
30        self(self, 2 * p + 1, mid, r);
31        pull(p);
32    };
33
34    build(build, 1, 0, n);
35 }
36
37 void pull(int p) {
38     info[p] = Mono::op(info[2 * p], info[2 * p + 1]);
39 }
40
41 void apply(int p, int len, const Y& v) {
42     info[p] = ActMono::act(info[p], v, len);
43     tag[p] = Tag::op(tag[p], v);
44 }
45
46 void push(int p, int len) {
47     int llen = len / 2;
48     apply(2 * p, llen, tag[p]);
49     apply(2 * p + 1, len - llen, tag[p]);
50     tag[p] = Tag::e();
51 }
52
53 void modify(int p, int l, int r, int x, const X& v) {
54     if (r - l == 1) {
55         info[p] = v;
56         return;
57     }
58     int mid = (l + r) / 2;
59     push(p, r - l);
60     if (x < mid) {
61         modify(2 * p, l, mid, x, v);
62     } else {
63         modify(2 * p + 1, mid, r, x, v);
64     }
65     pull(p);
66 }
67
68 void modify(int p, const X& v) {
69     modify(1, 0, n, p, v);
70 }

```

```

71 X rangeQuery(int p, int l, int r, int x, int y) {
72     if (y <= 1 || r <= x) {
73         return Mono::e();
74     }
75     if (x <= 1 && r <= y) {
76         return info[p];
77     }
78     int mid = (l + r) / 2;
79     push(p, r - 1);
80     return Mono::op(rangeQuery(2 * p, l, mid, x, y), rangeQuery(2 * p + 1, mid,
r, x, y));
81 }
82 X rangeQuery(int l, int r) {
83     return rangeQuery(1, 0, n, l, r);
84 }
85
86 void rangeApply(int p, int l, int r, int x, int y, const Y& v) {
87     if (y <= 1 || r <= x) {
88         return;
89     }
90     if (x <= 1 && r <= y) {
91         apply(p, r - 1, v);
92         return;
93     }
94     int mid = (l + r) / 2;
95     push(p, r - 1);
96     rangeApply(2 * p, l, mid, x, y, v);
97     rangeApply(2 * p + 1, mid, r, x, y, v);
98     pull(p);
99 }
100 void rangeApply(int l, int r, const Y& v) {
101     return rangeApply(1, 0, n, l, r, v);
102 }
103
104 template <typename F>
105 int findFirst(int p, int l, int r, int x, int y, F&& pred) {
106     if (y <= 1 || r <= x) {
107         return -1;
108     }
109     if (x <= 1 && r <= y && !pred(info[p])) {
110         return -1;
111     }
112     if (r - l == 1) {

```

```

113         return l;
114     }
115     int mid = (l + r) / 2;
116     push(p, r - 1);
117     int res = findFirst(2 * p, l, mid, x, y, pred);
118     if (res == -1) {
119         res = findFirst(2 * p + 1, mid, r, x, y, pred);
120     }
121     return res;
122 }
123 template <typename F>
124 int findFirst(int l, int r, F&& pred) {
125     return findFirst(1, 0, n, l, r, pred);
126 }
127
128 template <typename F>
129 int findLast(int p, int l, int r, int x, int y, F&& pred) {
130     if (y <= 1 || r <= x) {
131         return -1;
132     }
133     if (x <= 1 && r <= y && !pred(info[p])) {
134         return -1;
135     }
136     if (r - l == 1) {
137         return l;
138     }
139     int mid = (l + r) / 2;
140     push(p, r - 1);
141     int res = findLast(2 * p + 1, mid, r, x, y, pred);
142     if (res == -1) {
143         res = findLast(2 * p, l, mid, x, y, pred);
144     }
145     return res;
146 }
147 template <typename F>
148 int findLast(int l, int r, F&& pred) {
149     return findLast(1, 0, n, l, r, pred);
150 }
151 };

```

7.1.3 动态开点线段树

```

1 template <typename ActMono>

```

```

2 struct DynamicSegmentTree {
3     using Mono = typename ActMono::Mono;
4     using Tag = typename ActMono::Tag;
5     using X = typename ActMono::X;
6     using Y = typename ActMono::Y;
7
8     int n;
9     int root;
10    vector<X> info;
11    vector<Y> tag;
12    vector<int> lc, rc;
13
14    DynamicSegmentTree() : n(0), root(-1) {}
15    DynamicSegmentTree(int n_) {
16        init(n_);
17    }
18
19    void init(int n_) {
20        n = n_;
21        info.clear();
22        tag.clear();
23        lc.clear();
24        rc.clear();
25        root = newNode();
26    }
27
28    int newNode() {
29        info.push_back(Mono::e());
30        tag.push_back(Tag::e());
31        lc.push_back(-1);
32        rc.push_back(-1);
33        return int(info.size()) - 1;
34    }
35
36    void pull(int p) {
37        X left = (lc[p] != -1 ? info[lc[p]] : Mono::e());
38        X right = (rc[p] != -1 ? info[rc[p]] : Mono::e());
39        info[p] = Mono::op(left, right);
40    }
41
42    void apply(int p, int len, const Y& v) {
43        info[p] = ActMono::act(info[p], v, len);
44        tag[p] = Tag::op(tag[p], v);
45    }

```

```

46
47    void push(int p, int len) {
48        if (lc[p] == -1) {
49            lc[p] = newNode();
50        }
51        if (rc[p] == -1) {
52            rc[p] = newNode();
53        }
54        int llen = len / 2;
55        apply(lc[p], llen, tag[p]);
56        apply(rc[p], len - llen, tag[p]);
57        tag[p] = Tag::e();
58    }
59
60    void modify(int p, int l, int r, int x, const X& v) {
61        if (r - l == 1) {
62            info[p] = v;
63            return;
64        }
65        int mid = (l + r) / 2;
66        push(p, r - l);
67        if (x < mid) {
68            modify(lc[p], l, mid, x, v);
69        } else {
70            modify(rc[p], mid, r, x, v);
71        }
72        pull(p);
73    }
74
75    void modify(int x, const X& v) {
76        modify(root, 0, n, x, v);
77    }
78
79    X rangeQuery(int p, int l, int r, int x, int y) {
80        if (p == -1 || y <= l || r <= x) {
81            return Mono::e();
82        }
83        if (x <= l && r <= y) {
84            return info[p];
85        }
86        int mid = (l + r) / 2;
87        push(p, r - l);
88        return Mono::op(rangeQuery(lc[p], l, mid, x, y), rangeQuery(rc[p], mid, r,
89    x, y));

```

```

88 }
89 X rangeQuery(int l, int r) {
90     return rangeQuery(root, 0, n, l, r);
91 }
92
93 void rangeApply(int p, int l, int r, int x, int y, const Y& v) {
94     if (y <= l || r <= x) {
95         return;
96     }
97     if (x <= l && r <= y) {
98         apply(p, r - l, v);
99         return;
100     }
101     int mid = (l + r) / 2;
102     push(p, r - l);
103     rangeApply(lc[p], l, mid, x, y, v);
104     rangeApply(rc[p], mid, r, x, y, v);
105     pull(p);
106 }
107 void rangeApply(int l, int r, const Y& v) {
108     rangeApply(root, 0, n, l, r, v);
109 }
110
111 template <typename F>
112 int findFirst(int p, int l, int r, int x, int y, F&& pred) {
113     if (p == -1 || y <= l || r <= x) {
114         return -1;
115     }
116     if (x <= l && r <= y && !pred(info[p])) {
117         return -1;
118     }
119     if (r - l == 1) {
120         return l;
121     }
122     int mid = (l + r) / 2;
123     push(p, r - l);
124     int res = findFirst(lc[p], l, mid, x, y, pred);
125     if (res == -1) {
126         res = findFirst(rc[p], mid, r, x, y, pred);
127     }
128     return res;
129 }
130 template <typename F>

```

```

131 int findFirst(int l, int r, F&& pred) {
132     return findFirst(root, 0, n, l, r, pred);
133 }
134
135 template <typename F>
136 int findLast(int p, int l, int r, int x, int y, F&& pred) {
137     if (p == -1 || y <= l || r <= x) {
138         return -1;
139     }
140     if (x <= l && r <= y && !pred(info[p])) {
141         return -1;
142     }
143     if (r - l == 1) {
144         return l;
145     }
146     int mid = (l + r) / 2;
147     push(p, r - l);
148     int res = findLast(rc[p], mid, r, x, y, pred);
149     if (res == -1) {
150         res = findLast(lc[p], l, mid, x, y, pred);
151     }
152     return res;
153 }
154 template <typename F>
155 int findLast(int l, int r, F&& pred) {
156     return findLast(root, 0, n, l, r, pred);
157 }
158 };

```

7.2 树状数组

7.2.1 树状数组

```

1 template <typename Mono>
2 struct Fenwick {
3     using X = typename Mono::Type;
4
5     int n;
6     vector<X> a;
7
8     Fenwick() : n(0) {}
9     Fenwick(int n_) {
10         init(n_);
11     }
12

```

```

13 void init(int n_) {
14     n = n_;
15     a.assign(n, Mono::e());
16 }
17
18 void add(int x, const X& v) {
19     assert(0 <= x && x < n);
20     for (int i = x + 1; i <= n; i += i & -i) {
21         a[i - 1] = Mono::op(a[i - 1], v);
22     }
23 }
24
25 X sum(int x) {
26     assert(0 <= x && x <= n);
27     X res = Mono::e();
28     for (int i = x; i > 0; i -= i & -i) {
29         res = Mono::op(a[i - 1], res);
30     }
31     return res;
32 }
33
34 X rangeSum(int l, int r) {
35     assert(l <= r);
36     return Mono::op(Mono::inv(sum(l)), sum(r));
37 }
38
39 // lower_bound
40 int kth(X k) {
41     int x = 0;
42     for (int i = 1 << __lg(n); i; i /= 2) {
43         if (x + i <= n && k >= a[x + i - 1]) {
44             x += i;
45             k -= a[x - 1];
46         }
47     }
48     return x;
49 }
50 };

```

7.2.2 区间加、单点查树状数组

```

1 template <typename Abel>
2 struct DualFenwick {
3     using X = typename Abel::Type;

```

```

4
5     int n;
6     Fenwick<Abel> fen;
7
8     DualFenwick() : n(0), fen(0) {}
9     DualFenwick(int n_) {
10         init(n_);
11     }
12
13     void init(int n_) {
14         n = n_;
15         fen.init(n);
16     }
17
18     void add(int l, int r, const X& v) {
19         assert(0 <= l && l <= r && r <= n);
20         fen.add(l, v);
21         if (r < n) {
22             fen.add(r, Abel::inv(v));
23         }
24     }
25
26     X query(int x) {
27         assert(0 <= x && x < n);
28         return fen.sum(x + 1);
29     }
30 };

```

7.2.3 区间加、区间和树状数组

```

1 template <typename Abel>
2 struct FenwickRangeAdd {
3     using X = typename Abel::Type;
4
5     int n;
6     Fenwick<Abel> fen0, fen1;
7
8     FenwickRangeAdd() : n(0), fen0(1), fen1(1) {}
9     FenwickRangeAdd(int n_) {
10         init(n_);
11     }
12
13     void init(int n_) {
14         n = n_;
15         fen0.init(n + 1);

```

```

16     fen1.init(n + 1);
17 }
18
19 void add(int l, int r, const X& v) {
20     assert(0 <= l && l <= r && r <= n);
21     fen0.add(l, Abel::power(v, -static_cast<i64>(l)));
22     fen0.add(r, Abel::power(v, static_cast<i64>(r)));
23     fen1.add(l, v);
24     fen1.add(r, Abel::inv(v));
25 }
26
27 X sum(int r) {
28     assert(0 <= r && r <= n);
29     return Abel::op(Abel::power(fen1.sum(r), static_cast<i64>(r)),
30 fen0.sum(r));
31 }
32
33 X rangeSum(int l, int r) {
34     assert(0 <= l && l <= r && r <= n);
35     return Abel::op(Abel::inv(sum(l)), sum(r));
36 }
37 };

```

7.2.4 二维树状数组

```

1 template <typename Mono>
2 struct Fenwick2D {
3     using X = typename Mono::Type;
4
5     int h, w;
6     vector<X> a;
7
8     Fenwick2D() : h(0), w(0) {}
9     Fenwick2D(int h_, int w_) {
10         init(h_, w_);
11     }
12
13     void init(int h_, int w_) {
14         h = h_;
15         w = w_;
16         a.assign((h + 1) * (w + 1), Mono::e());
17     }
18
19     void add(int x, int y, const X& v) {
20         assert(0 <= x && x < h);

```

```

21         assert(0 <= y && y < w);
22         for (int i = x + 1; i <= h; i += i & -i) {
23             for (int j = y + 1; j <= w; j += j & -j) {
24                 a[i * (w + 1) + j] = Mono::op(a[i * (w + 1) + j], v);
25             }
26         }
27     }
28
29     X sum(int x, int y) {
30         assert(0 <= x && x <= h);
31         assert(0 <= y && y <= w);
32         X res = Mono::e();
33         for (int i = x; i > 0; i -= i & -i) {
34             for (int j = y; j > 0; j -= j & -j) {
35                 res = Mono::op(res, a[i * (w + 1) + j]);
36             }
37         }
38         return res;
39     }
40
41     X rangeSum(int x1, int x2, int y1, int y2) {
42         assert(0 <= x1 && x1 <= x2 && x2 <= h);
43         assert(0 <= y1 && y1 <= y2 && y2 <= w);
44         X s22 = sum(x2, y2);
45         X s12 = sum(x1, y2);
46         X s21 = sum(x2, y1);
47         X s11 = sum(x1, y1);
48         return Mono::op(Mono::op(s22, Mono::inv(s12)), Mono::op(Mono::inv(s21),
49 s11));
50     }
51 };

```

7.3 倍增

```

1 template <typename Mono>
2 struct BinaryLifting {
3     using X = typename Mono::Type;
4
5     int n, logn;
6     vector<vector<int>>> nxt;
7     vector<vector<X>>> info;
8
9     BinaryLifting() : n(0) {}
10    BinaryLifting(int n_) {

```

```

11     init(n_);
12 }
13
14 void init(int n_) {
15     n = n_;
16     logn = __lg(n);
17     nxt.assign(logn + 1, vector<int>(n, -1));
18     info.assign(logn + 1, vector<X>(n, Mono::e()));
19 }
20
21 void add(int u, int v, const X& x) {
22     assert(-1 <= v && v < n);
23     nxt[0][u] = v;
24     info[0][u] = x;
25 }
26
27 void build() {
28     for (int k = 0; k < logn; k++) {
29         for (int u = 0; u < n; u++) {
30             int v = nxt[k][u];
31             if (v == -1) {
32                 nxt[k + 1][u] = -1;
33                 info[k + 1][u] = info[k][u];
34             } else {
35                 nxt[k + 1][u] = nxt[k][v];
36                 info[k + 1][u] = Mono::op(info[k][u], info[k][v]);
37             }
38         }
39     }
40 }
41
42 pair<int, X> query(int u, i64 step) {
43     assert(0 <= step && step < (1LL << (logn + 1)));
44     X res = Mono::e();
45     for (int k = logn; k >= 0 && u != -1; k--) {
46         if ((step >> k) & 1) {
47             res = Mono::op(res, info[k][u]);
48             u = nxt[k][u];
49         }
50     }
51     return {u, res};
52 }
53

```

```

54 int jump(int u, i64 step) {
55     assert(0 <= step && step < (1LL << (logn + 1)));
56     for (int k = logn; k >= 0 && u != -1; k--) {
57         if ((step >> k) & 1) {
58             u = nxt[k][u];
59         }
60     }
61     return u;
62 }
63 };

```

7.4 并查集

```

1 struct DSU {
2     vector<int> f, siz;
3
4     DSU() {}
5     DSU(int n) {
6         init(n);
7     }
8
9     void init(int n) {
10         f.resize(n);
11         iota(f.begin(), f.end(), 0);
12         siz.assign(n, 1);
13     }
14
15     int find(int x) {
16         while (x != f[x]) {
17             x = f[x] = f[f[x]];
18         }
19         return x;
20     }
21
22     bool same(int x, int y) {
23         return find(x) == find(y);
24     }
25
26     bool merge(int x, int y) {
27         x = find(x);
28         y = find(y);
29         if (x == y) {
30             return false;
31         }

```

```

32     siz[x] += siz[y];
33     f[y] = x;
34     return true;
35 }
36
37 int size(int x) {
38     return siz[find(x)];
39 }
40 };

```

7.5 稀疏表

```

1  template <typename Mono>
2  struct SparseTable {
3      using X = typename Mono::Type;
4
5      int n;
6      vector<vector<X>> st;
7
8      SparseTable(const vector<X>& a) {
9          init(a);
10     }
11
12     void init(const vector<X>& a) {
13         n = a.size();
14         int logn = __lg(n);
15         st.assign(logn + 1, vector<X>(n));
16         for (int i = 0; i < n; i++) {
17             st[0][i] = a[i];
18         }
19         for (int j = 1; (1 << j) <= n; j++) {
20             for (int i = 0; i + (1 << j) <= n; i++) {
21                 st[j][i] = Mono::op(st[j - 1][i], st[j - 1][i + (1 << (j - 1))]);
22             }
23         }
24     }
25
26     X query(int l, int r) {
27         assert(l < r);
28         int len = r - l;
29         int t = __lg(len);
30         return Mono::op(st[t][l], st[t][r - (1 << t)]);
31     }
32 };

```

7.6 01-Trie

```

1  struct BinaryTrie {
2      struct Node {
3          array<int, 2> ch;
4          int end;
5          Node() {
6              ch[0] = ch[1] = 0;
7              end = 0;
8          }
9      };
10
11     vector<Node> tr;
12     int tot;
13
14     BinaryTrie(int n) {
15         init(n);
16     }
17
18     void init(int n) {
19         tr.resize(n);
20         tot = 1;
21     }
22
23     void insert(int x) {
24         int p = 1;
25         for (int i = 30; i >= 0; i--) {
26             int c = x >> i & 1;
27             if (!tr[p].ch[c]) {
28                 ++tot;
29                 tr[p].ch[c] = tot;
30             }
31             p = tr[p].ch[c];
32             tr[p].end++;
33         }
34     }
35
36     int query(int x) {
37         int p = 1, res = 0;
38         for (int i = 30; i >= 0; i--) {
39             int c = x >> i & 1;
40             if (tr[p].ch[c ^ 1]) {
41                 res |= (1 << i);
42                 p = tr[p].ch[c ^ 1];

```

```

43         } else {
44             p = tr[p].ch[c];
45         }
46     }
47     return res;
48 }
49
50 void erase(int x) {
51     int p = 1;
52     for (int i = 30; i >= 0; i--) {
53         int c = x >> i & 1;
54         int nxt = tr[p].ch[c];
55         tr[nxt].end--;
56         if (tr[nxt].end == 0) {
57             tr[p].ch[c] = 0;
58         }
59         p = nxt;
60     }
61 }
62 };
63

```

7.7 笛卡尔树

```

1 template <typename T, bool IS_MIN = false>
2 struct CartesianTree {
3     int n;
4     vector<T>& a;
5     vector<int> L, R, p;
6     // range[i]: the interval of the subtree rooted at i
7     vector<pair<int, int>> range;
8     int root;
9
10    CartesianTree(vector<T>& a_) : n(a_.size()), a(a_), root(-1) {
11        L.assign(n, -1);
12        R.assign(n, -1);
13        p.assign(n, -1);
14        range.assign(n, {0, n});
15        vector<int> stk;
16
17        auto cmp = [&](int i, int j) -> bool {
18            if (a[i] == a[j]) {
19                return i < j;
20            }
21        }
22    };
23

```

```

21         if constexpr (IS_MIN) {
22             return a[i] < a[j];
23         } else {
24             return a[j] < a[i];
25         }
26     };
27
28     for (int i = 0; i < n; i++) {
29         while (!stk.empty() && cmp(i, stk.back())) {
30             L[i] = stk.back();
31             stk.pop_back();
32         }
33         range[i].first = stk.empty() ? 0 : stk.back() + 1;
34         if (!stk.empty()) {
35             R[stk.back()] = i;
36         }
37         stk.push_back(i);
38     }
39     if (!stk.empty()) {
40         root = stk.front();
41     }
42     stk.clear();
43     for (int i = n - 1; i >= 0; i--) {
44         while (!stk.empty() && cmp(i, stk.back())) {
45             stk.pop_back();
46         }
47         range[i].second = stk.empty() ? n : stk.back();
48         stk.push_back(i);
49     }
50     for (int i = 0; i < n; i++) {
51         if (L[i] != -1) {
52             p[L[i]] = i;
53         }
54         if (R[i] != -1) {
55             p[R[i]] = i;
56         }
57     }
58 }
59 };

```

8 线性代数

8.1 矩阵

```

1 template <typename T>
2 struct Matrix {
3     int n, m;
4     vector<vector<T>> a;
5
6     Matrix() : n(0), m(0) {}
7     Matrix(int n_, int m_, T v = T{}) : n(n_), m(m_), a(n, vector<T>(m, v)) {}
8     Matrix(int n_) : Matrix(n_, n_) {}
9
10    static Matrix identity(int n) {
11        Matrix res(n);
12        for (int i = 0; i < n; i++) {
13            res[i][i] = T{1};
14        }
15        return res;
16    }
17
18    vector<T>& operator[](int i) {
19        return a[i];
20    }
21    const vector<T>& operator[](int i) const {
22        return a[i];
23    }
24 };
25
26 template <typename T>
27 Matrix<T> operator*(const Matrix<T>& a, const Matrix<T>& b) {
28     assert(a.m == b.n);
29     Matrix<T> c(a.n, b.m);
30     for (int i = 0; i < c.n; i++) {
31         for (int k = 0; k < a.m; k++) {
32             for (int j = 0; j < c.m; j++) {
33                 c.a[i][j] += a.a[i][k] * b.a[k][j];
34             }
35         }
36     }
37     return c;
38 }
39
40 template <typename T>
41 Matrix<T>& operator*=(Matrix<T>& a, const Matrix<T>& b) {

```

```

42     return a = a * b;
43 }
44
45 template <typename T, integral U>
46 Matrix<T> power(Matrix<T> a, U b) {
47     Matrix<T> res = Matrix<T>::identity(a.n);
48     for (; b > 0; b >>= 1) {
49         if (b & 1) {
50             res *= a;
51         }
52         a *= a;
53     }
54     return res;
55 }
56
57 template <typename T>
58 Matrix<T> transpose(const Matrix<T>& a) {
59     Matrix<T> res(a.m, a.n);
60     for (int i = 0; i < a.n; i++) {
61         for (int j = 0; j < a.m; j++) {
62             res[j][i] = a[i][j];
63         }
64     }
65     return res;
66 }
67
68 template <typename T>
69 Matrix<T> augment(const Matrix<T>& a, const Matrix<T>& b) {
70     assert(a.n == b.n);
71     Matrix<T> res(a.n, a.m + b.m);
72     for (int i = 0; i < a.n; i++) {
73         for (int j = 0; j < a.m; j++) {
74             res[i][j] = a[i][j];
75         }
76         for (int j = 0; j < b.m; j++) {
77             res[i][a.m + j] = b[i][j];
78         }
79     }
80     return res;
81 }
82
83 template <typename T>
84 Matrix<T> augment(const Matrix<T>& a, const vector<T>& b) {

```

```

85  assert(a.n == int(b.size()));
86  Matrix<T> res(a.n, a.m + 1);
87  for (int i = 0; i < a.n; i++) {
88      for (int j = 0; j < a.m; j++) {
89          res[i][j] = a[i][j];
90      }
91      res[i][a.m] = b[i];
92  }
93  return res;
94  }

```

8.2 高斯消元

```

1  template <typename T>
2  pair<vector<int>, T> eliminate(Matrix<T>& A, int lim, bool full) {
3      // full == true -> RREF
4      // full == false -> REF
5      vector<int> pivot;
6      T prod = T{1};
7      int r = 0;
8      for (int j = 0; j < lim && r < A.n; j++) {
9          int p = r;
10         while (p < A.n && A[p][j] == T{0}) {
11             p++;
12         }
13         if (p == A.n) {
14             continue;
15         }
16         if (p != r) {
17             swap(A[p], A[r]);
18             prod = -prod;
19         }
20         T x = A[r][j];
21         prod *= x;
22         pivot.push_back(j);
23         T inv = T{1} / x;
24         for (int k = j; k < A.m; k++) {
25             A[r][k] *= inv;
26         }
27         for (int i = 0; i < A.n; i++) {
28             if (i == r || (!full && i < r)) {
29                 continue;
30             }

```

```

31         T c = A[i][j];
32         if (c == T{0}) {
33             continue;
34         }
35         A[i][j] = T{0};
36         for (int k = j + 1; k < A.m; k++) {
37             A[i][k] -= c * A[r][k];
38         }
39     }
40     r++;
41 }
42 return {pivot, prod};
43 }
44
45 template <typename T>
46 pair<vector<int>, T> gauss(Matrix<T>& A) {
47     return eliminate(A, A.m, false);
48 }
49 template <typename T>
50 pair<vector<int>, T> gaussJordan(Matrix<T>& A, int lim) {
51     return eliminate(A, lim, true);
52 }
53
54 template <typename T>
55 pair<T, Matrix<T>> matrixInv(const Matrix<T>& a) {
56     if (a.n != a.m) {
57         return {T{0}, {}};
58     }
59     int n = a.n;
60     auto aug = augment(a, Matrix<T>::identity(n));
61     auto [pivot, det] = gaussJordan(aug, n);
62     if (pivot.size() < n) {
63         return {T{0}, {}};
64     }
65     Matrix<T> inv(n);
66     for (int i = 0; i < n; i++) {
67         for (int j = 0; j < n; j++) {
68             inv[i][j] = aug[i][n + j];
69         }
70     }
71     return {det, inv};
72 }
73

```

```

74 template <typename T>
75 int matrixRank(Matrix<T> a) {
76     auto [pivot, prod] = gauss(a);
77     return pivot.size();
78 }
79
80 template <typename T>
81 T determinant(Matrix<T> A) {
82     assert(A.n == A.m);
83     auto [pivot, det] = gauss(A);
84     if (pivot.size() < A.n) {
85         return T{0};
86     }
87     return det;
88 }
89
90 template <typename T>
91 vector<vector<T>> solveLinear(const Matrix<T>& A, const vector<T>& b) {
92     int n = A.n;
93     int m = A.m;
94     auto aug = augment(A, b);
95     auto [pivot, prod] = gaussJordan(aug, m);
96     int r = pivot.size();
97     for (int i = r; i < n; i++) {
98         if (aug[i][m] != T{0}) {
99             return {};
100         }
101     }
102     vector<int> row(m, -1);
103     for (int i = 0; i < r; i++) {
104         row[pivot[i]] = i;
105     }
106     vector<vector<T>> sol(1, vector<T>(m));
107     for (int j = 0; j < m; j++) {
108         if (row[j] != -1) {
109             sol[0][j] = aug[row[j]][m];
110         }
111     }
112     for (int j = 0; j < m; j++) {
113         if (row[j] != -1) {
114             continue;
115         }
116         vector<T> x(m);

```

```

117         x[j] = T{-1};
118         for (int i = 0; i < r; i++) {
119             x[pivot[i]] = aug[i][j];
120         }
121         sol.push_back(x);
122     }
123     return sol;
124 }

```

8.3 GF(2) 空间

8.3.1 异或线性基

```

1 struct Basis {
2     array<u64, 64> p{};
3     bool norm = false;
4
5     int size() {
6         int r = 0;
7         for (int i = 0; i < 64; i++) {
8             r += p[i] != 0;
9         }
10        return r;
11    }
12
13    bool insert(u64 x) {
14        for (int i = 63; i >= 0; i--) {
15            if ((x >> i & 1)) {
16                continue;
17            }
18            if (!p[i]) {
19                p[i] = x;
20                norm = false;
21                return true;
22            }
23            x ^= p[i];
24        }
25        return false;
26    }
27
28    bool contain(u64 x) {
29        for (int i = 63; i >= 0; i--) {
30            if (x >> i & 1) {
31                x ^= p[i];
32            }

```

```

33     }
34     return x == 0;
35 }
36
37 u64 getMax(u64 x = 0) {
38     for (int i = 63; i >= 0; i--) {
39         if ((x ^ p[i]) > x) {
40             x ^= p[i];
41         }
42     }
43     return x;
44 }
45
46 u64 getMin(u64 x = 0) {
47     for (int i = 63; i >= 0; i--) {
48         if ((x ^ p[i]) < x) {
49             x ^= p[i];
50         }
51     }
52     return x;
53 }
54
55 void normalize() {
56     if (norm == true) {
57         return;
58     }
59     norm = true;
60     for (int i = 63; i >= 0; i--) {
61         if (!p[i]) {
62             continue;
63         }
64         for (int j = i - 1; j >= 0; j--) {
65             if (p[i] >> j & 1) {
66                 p[i] ^= p[j];
67             }
68         }
69     }
70 }
71
72 u64 kth(u64 k) {
73     normalize();
74     vector<u64> b;
75     for (int i = 0; i < 64; i++) {

```

```

76         if (p[i]) {
77             b.push_back(p[i]);
78         }
79     }
80     int r = b.size();
81     if (r < 64) {
82         if (k < (1LL << r)) {
83             return -1;
84         }
85     }
86     u64 res = 0;
87     for (int i = 0; i < r; i++) {
88         if (k >> i & 1) {
89             res ^= b[i];
90         }
91     }
92     return res;
93 }
94 };

```

8.3.2 逆矩阵

```

1 namespace gf2 {
2
3 template <typename U>
4 vector<U> inverse(int n, vector<U> a) {
5     vector<U> b(n);
6     for (int i = 0; i < n; i++) {
7         b[i] = U{1} << i;
8     }
9     for (int j = 0; j < n; j++) {
10         int p = j;
11         while (p < n && !(a[p] >> j & U{1})) {
12             p++;
13         }
14         if (p == n) {
15             return {};
16         }
17         swap(a[p], a[j]);
18         swap(b[p], b[j]);
19         for (int i = 0; i < n; i++) {
20             if (i != j && (a[i] >> j & U{1})) {
21                 a[i] ^= a[j];

```

```

22         b[i] ^= b[j];
23     }
24 }
25 }
26 return b;
27 }
28 }
29 }

```

8.3.3 转置

```

1 namespace gf2 {
2
3 template <typename U>
4 vector<U> transpose(int n, int m, const vector<U>& a) {
5     vector<U> b(m);
6     for (int i = 0; i < n; i++) {
7         for (int j = 0; j < m; j++) {
8             if (a[i] >> j & U{1}) {
9                 b[j] |= U{1} << i;
10            }
11        }
12    }
13    return b;
14 }
15 }
16 }

```

8.3.4 GF(2) 线性方程组

```

1 namespace gf2 {
2
3 template <unsigned_integral U>
4 vector<U> solveLinear(int n, int m, vector<U> a, U b) {
5     vector<U> mat(m);
6     U rhs = 0;
7     for (int i = 0; i < n; i++) {
8         U x = a[i], c = b >> i & 1;
9         for (int j = m - 1; j >= 0; j--) {
10             if ((x ^ mat[j]) < x) {
11                 x ^= mat[j];
12                 c ^= rhs >> j & 1;
13            }
14        }
15        if (!x) {

```

```

16            if (c) {
17                return {};
18            }
19            continue;
20        }
21        int k = bit_width(x) - 1;
22        for (int j = k; j < m; j++) {
23            if ((mat[j] ^ x) < mat[j]) {
24                mat[j] ^= x;
25                rhs ^= c << j;
26            }
27        }
28        mat[k] = x;
29        rhs |= c << k;
30    }
31    vector<U> res{rhs};
32    mat = transpose(m, m, mat);
33    for (int j = 0; j < m; j++) {
34        if (!(mat[j] >> j & 1)) {
35            res.push_back(mat[j] | (U{1} << j));
36        }
37    }
38    return res;
39 }
40 }
41 }

```

9 字符串

9.1 Manacher

```

1 vector<int> manacher(string s) {
2     string t = "#";
3     for (auto c : s) {
4         t += c;
5         t += '#';
6     }
7     int n = t.size();
8     vector<int> r(n);
9     for (int i = 0, j = 0; i < n; i++) {
10        if (2 * j - i >= 0 && j + r[j] > i) {
11            r[i] = min(r[2 * j - i], j + r[j] - i);
12        }
13        while (i - r[i] >= 0 && i + r[i] < n && t[i - r[i]] == t[i + r[i]]) {

```

```

14         r[i] += 1;
15     }
16     if (i + r[i] > j + r[j]) {
17         j = i;
18     }
19 }
20 return r;
21 }

```

9.2 KMP

```

1 vector<int> kmp(string s) {
2     int n = s.size();
3     vector<int> f(n + 1);
4     for (int i = 1, j = 0; i < n; i++) {
5         while (j && s[i] != s[j]) {
6             j = f[j];
7         }
8         j += (s[i] == s[j]);
9         f[i + 1] = j;
10    }
11    return f;
12 }
13

```

9.3 Trie

```

1 struct Trie {
2     struct Node {
3         array<int, 26> nxt;
4         int pass, end;
5
6         Node() {
7             nxt.fill(-1);
8             end = 0;
9             pass = 0;
10        }
11    };
12
13    vector<Node> tr;
14    int tot;
15
16    Trie(int n) {
17        init(n);
18    }
19

```

```

20 void init(int n) {
21     tr.resize(n);
22     tot = 0;
23 }
24
25 void insert(string& s) {
26     int p = 0;
27     ++tr[p].pass;
28     for (char& ch : s) {
29         int c = ch - 'a';
30         if (tr[p].nxt[c] == -1) {
31             ++tot;
32             tr[p].nxt[c] = tot;
33         }
34         p = tr[p].nxt[c];
35         ++tr[p].pass;
36     }
37     ++tr[p].end;
38 }
39
40 int count(string& s) {
41     int p = 0;
42     for (char& ch : s) {
43         int c = ch - 'a';
44         if (tr[p].nxt[c] == -1) {
45             return false;
46         }
47         p = tr[p].nxt[c];
48     }
49     return tr[p].end;
50 }
51 };
52

```

9.4 Z 函数

```

1 vector<int> Z(string s) {
2     int n = s.size();
3     vector<int> z(n + 1);
4     z[0] = n;
5     for (int i = 1, j = 1; i < n; i++) {
6         z[i] = max(0, min(j + z[j] - i, z[i - j]));
7         while (i + z[i] < n && s[z[i]] == s[i + z[i]]) {
8             z[i]++;
9         }
10    }
11 }

```

```

9         }
10         if (i + z[i] > j + z[j]) {
11             j = i;
12         }
13     }
14     return z;
15 }
16

```

9.5 AC 自动机

```

1 struct AhoCorasick {
2     static constexpr int ALPHABET = 26;
3     struct Node {
4         int len;
5         int link;
6         array<int, ALPHABET> next;
7         Node() : len{0}, link{0}, next{} {}
8     };
9
10    vector<Node> t;
11
12    AhoCorasick() {
13        init();
14    }
15
16    void init() {
17        t.assign(2, Node());
18        t[0].next.fill(1);
19        t[0].len = -1;
20    }
21
22    int newNode() {
23        t.emplace_back();
24        return t.size() - 1;
25    }
26
27    int add(const string &a) {
28        int p = 1;
29        for (auto c : a) {
30            int x = c - 'a';
31            if (t[p].next[x] == 0) {
32                t[p].next[x] = newNode();
33                t[t[p].next[x]].len = t[p].len + 1;

```

```

34        }
35        p = t[p].next[x];
36    }
37    return p;
38 }
39
40 void work() {
41     queue<int> q;
42     q.push(1);
43
44     while (!q.empty()) {
45         int x = q.front();
46         q.pop();
47
48         for (int i = 0; i < ALPHABET; i++) {
49             if (t[x].next[i] == 0) {
50                 t[x].next[i] = t[t[x].link].next[i];
51             } else {
52                 t[t[x].next[i]].link = t[t[x].link].next[i];
53                 q.push(t[x].next[i]);
54             }
55         }
56     }
57 }
58
59 int next(int p, int x) {
60     return t[p].next[x];
61 }
62
63 int link(int p) {
64     return t[p].link;
65 }
66
67 int len(int p) {
68     return t[p].len;
69 }
70
71 int size() {
72     return t.size();
73 }
74 };
75

```

10 杂项

10.1 debug

```

1 using namespace std;
2
3 #define dbg(...) { cerr << "[" << __LINE__ << "]" "; pr(__VA_ARGS__); cerr <<
  '\n'; }
4
5 void pr(bool b) { cerr << (b ? "T" : "F"); }
6 void pr(char c) { cerr << "'" << c << "'"; }
7 void pr(string s) { cerr << "'" << s << "'"; }
8 void pr(const char* s) { cerr << "'" << s << "'"; }
9
10 template <class T>
11 void pr(T&& x) {
12     using U = remove_reference_t<T>;
13     if constexpr (requires { begin(x); }) {
14         bool br = requires { typename U::key_type; };
15         cerr << (br ? '{' : '[');
16         bool f = 0;
17         for (auto&& e : x) {
18             if (exchange(f, 1)) cerr << ",";
19             if constexpr (requires { typename U::mapped_type; }) { pr(e.first);
20                 cerr << ":"; pr(e.second); }
21             else pr(e);
22         }
23         cerr << (br ? '}' : ']');
24     } else if constexpr (requires { x.first; x.second; }) {
25         cerr << '('; pr(x.first); cerr << ","; pr(x.second); cerr << ')';
26     } else if constexpr (requires { get<0>(x); }) {
27         cerr << '(';
28         bool f = 0;
29         apply([&](auto... e) { ((cerr << (exchange(f, 1) ? "," : ""),
30             pr(e)), ...); }, x);
31         cerr << ')';
32     } else if constexpr (requires { x.top(); }) {
33         vector<typename U::value_type> v;
34         while (!x.empty()) v.push_back(x.top()), x.pop();
35         pr(v);
36     } else if constexpr (requires { x.front(); }) {
37         vector<typename U::value_type> v;
38         while (!x.empty()) v.push_back(x.front()), x.pop();
39         pr(v);
40     } else {

```

```

39         cerr << x;
40     }
41 }
42
43 template <class A, class... B> requires (sizeof...(B) > 0)
44 void pr(A a, B... b) { pr(a); ((cerr << " ", pr(b)), ...); }

```

10.2 template

```

1 using namespace std;
2
3 using i64 = int64_t;
4 using u64 = uint64_t;
5 using u32 = uint32_t;
6
7 using i128 = __int128_t;
8 using u128 = __uint128_t;
9
10 template <typename T>
11 constexpr T inf = numeric_limits<T>::max() / 2;
12 template <>
13 constexpr int inf<int> = 1e9;
14 template <>
15 constexpr i64 inf<i64> = 1e18;
16
17 void solve();
18
19 int main() {
20     ios::sync_with_stdio(false);
21     cin.tie(nullptr);
22
23     int t;
24     cin >> t;
25
26     while (t--) {
27         solve();
28     }
29
30     return 0;
31 }

```

10.3 二分答案

```

1 template <typename T, typename F>
2 T firstTrue(T lo, T hi, F&& pred) {
3     while (lo < hi) {

```

```

4     T mid = (lo + hi) / 2;
5     if (pred(mid)) {
6         hi = mid;
7     } else {
8         lo = mid + 1;
9     }
10 }
11 return lo;
12 }
13
14 template <typename T, typename F>
15 T lastTrue(T lo, T hi, F&& pred) {
16     while (hi - lo > 1) {
17         T mid = (lo + hi) / 2;
18         if (pred(mid)) {
19             lo = mid;
20         } else {
21             hi = mid;
22         }
23     }
24     return lo;
25 }

```

10.4 i128

```

1 using i128 = __int128;
2
3 ostream& operator<<(ostream& os, i128 n) {
4     if (n == 0) {
5         return os << 0;
6     }
7     string s;
8     while (n > 0) {
9         s += char('0' + n % 10);
10        n /= 10;
11    }
12    reverse(s.begin(), s.end());
13    return os << s;
14 }
15
16 i128 toi128(const string& s) {
17     i128 n = 0;
18     for (auto c : s) {
19         n = n * 10 + (c - '0');

```

```

20     }
21     return n;
22 }
23
24 i128 sqrti128(i128 n) {
25     i128 lo = 0, hi = 1E16;
26     while (lo < hi) {
27         i128 x = (lo + hi + 1) / 2;
28         if (x * x <= n) {
29             lo = x;
30         } else {
31             hi = x - 1;
32         }
33     }
34     return lo;
35 }
36
37 i128 gcd(i128 a, i128 b) {
38     while (b) {
39         a %= b;
40         swap(a, b);
41     }
42     return a;
43 }
44 }

```

10.5 日期

```

1 const int d[] = {31, 28, 31, 30, 31, 30, 31, 31, 30, 31, 30, 31};
2
3 bool isLeap(int y) {
4     return y % 400 == 0 || (y % 4 == 0 && y % 100 != 0);
5 }
6
7 int daysInMonth(int y, int m) {
8     return d[m - 1] + (isLeap(y) && m == 2);
9 }
10
11 int getDay(int y, int m, int d) {
12     int ans = 0;
13     for (int i = 1970; i < y; i++) {
14         ans += 365 + isLeap(i);
15     }
16     for (int i = 1; i < m; i++) {
17         ans += daysInMonth(y, i);

```

```
18     }  
19     ans += d;  
20     return (ans + 2) % 7 + 1;  
21 }  
22
```